The Impact of Improved Water and Sanitation on Child Nutritional Outcomes in Sub-Saharan Africa: Evidence from DHS and Geospatial Data

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Abstract

This thesis asks whether water and sanitation (WASH) conditions are associated with the anthropometric outcomes of children under five, measured as height-for-age (HAZ), weight-for-age (WAZ) and weight-for-height (WHZ) z-scores, in four Sub-Saharan African countries: Ethiopia, Ghana, Kenya and Nigeria. The baseline analysis distinguishes between a child's own household facilities and the WASH conditions of the surrounding community. The four countries were selected for data availability and recency, not to represent the region as a whole. Both analyses are cross-sectional and observational, and every estimate is reported as an association, not as a causal effect.

The baseline sample pools the most recent Demographic and Health Survey (DHS) round for each country (2022 to 2025). Household WASH is the improved or unimproved facility classification of the child's own household. Community WASH is a leave-one-out rate of improved-facility coverage among the other sampled households in the same DHS cluster, so a child's own household never enters its community measure. Each exposure is estimated under the fixed-effects structure its construction requires: cluster fixed effects for household WASH, which compare households within the same cluster, and region fixed effects for community WASH, which varies almost only between clusters. No household cluster-fixed-effects coefficient reaches significance at the 5% level for any of the three outcomes. Household water and HAZ comes closest ($p = .062$, crossing only a 10% threshold); for sanitation and HAZ, $p = .548$. These results are inconclusive rather than evidence that no association exists. Community coverage shows a large, significant association before covariate adjustment for most outcome and exposure pairs; the exception is water and WHZ, which is small and non-significant from the start. Once socioeconomic controls and region fixed effects are added, none of the community associations remains statistically significant, and the fully adjusted HAZ estimates for water ($p = .573$) and sanitation ($p = .894$) are both inconclusive. This attenuation is consistent with confounding by local socioeconomic conditions, but it does not establish confounding as the explanation.

A separate, preliminary extension uses an independently constructed sample: earlier DHS rounds for Ghana (2014), Kenya (2014) and Nigeria (2018), harmonised by IPUMS-DHS and linked to modelled annual gridded estimates of local improved water and sanitation coverage from the Institute for Health Metrics and Evaluation (IHME). It responds to one limitation of the baseline design, which records WASH once, at the survey date, rather than during each child's own early life. Within this sample, it asks whether modelled coverage during a child's own early-life window, here the first 12 postnatal months, is associated with later anthropo-

metric outcomes, using cluster fixed effects and country-specific birth-year terms. Ethiopia's 2016 round enters only as a provisional sensitivity analysis, because its interview-date conversion could not be validated. The exposure is an area average of modelled coverage around displaced cluster coordinates; it is neither observed household access nor population-weighted coverage. Because the historical data do not separate a child's household from its community, the extension does not repeat the household-versus-community comparison.

The selected primary finding is the pooled established-date association between local water coverage and HAZ: $\beta = 1.03$ per 10 percentage points of modelled coverage (95% CI [0.51, 1.54], $p < .001$). It is one of six nominally significant cells ($p < .05$) among the 24 primary cells, and no adjustment for multiple testing was applied. The estimate depends on the equal-country weighting convention: unweighted, on the same 23,018 observations, it is 0.41 (95% CI [0.03, 0.80], $p = .035$), and equal country weight totals do not give the three countries equal influence on the pooled slope. All 24 primary cells were reproduced with a second estimation library on a separately rebuilt sample; the other 267 of the 291 historical models passed internal checks and, apart from two unweighted counterparts, were not re-estimated with a second library. This reproduction confirms the computation, not causality or robustness to multiple testing. The models are weighted with the IPUMS-DHS PERWEIGHT variable. No official documentation of an anthropometry-specific weight for the measured subsample could be located, so the treatment of measurement selection rests on a stated working inference, and the estimates are weighted conditional associations among eligible measured children rather than population-representative estimates. The exposure source also draws on the Ghanaian and Kenyan DHS surveys themselves, and current residence stands in for early-life residence.

The baseline and historical samples differ at the same time in their children and survey rounds, exposure construction, eligibility rules and control sets, so the contrast between the largely null baseline results and the historical finding cannot be attributed to exposure timing. The historical result is a preliminary, qualified finding and does not overturn the more cautious baseline conclusions. The thesis contributes a side-by-side comparison of household and leave-one-out community WASH within one four-country sample and a preliminary, qualified extension that assigns modelled local coverage to each child's early-life window, each read with its own limitations. None of the results is evidence for or against the effectiveness of WASH investment as a policy instrument.

1 Introduction

Child stunting remains one of the most persistent markers of population health disadvantage in Sub-Saharan Africa. Linear growth faltering is concentrated in the first two years of life, is difficult to reverse once established, and its consequences extend well beyond anthropometry to cognitive development, school attainment, and adult earning capacity (Victora et al. 2008; Victora et al. 2010; Black et al. 2013). Among the environmental exposures proposed to explain why children in similar economic circumstances grow at such different rates, water, sanitation, and hygiene (WASH) conditions occupy a central place in both the public-health and the development-economics literature, via several biologically plausible mechanisms supported to varying degrees: inadequate water and sanitation can raise exposure to faecal pathogens, increasing diarrhoeal disease, soil-transmitted helminth infection, and sub-clinical environmental enteric dysfunction (EED), each of which could divert nutritional intake toward immune response rather than growth (Cumming and Cairncross 2016) — distinct, separately evidenced pathways, not one undisputed mechanism, with EED specifically flagged as the least settled and needing more research. Even the strongest recent causal evidence against basic WASH interventions reducing stunting does not dispute this plausibility (Section 2.3), but plausibility is not the same claim as an established causal mechanism, and this thesis does not conflate the two. What is considerably less settled is whether the mechanism translates into a *detectable, policy-relevant statistical association* in observational survey data once household wealth and local development are accounted for, and whether a *basic facility-level* WASH classification, as opposed to a comprehensive intervention package, is the right exposure in which to look for it — the narrower, empirical question this thesis addresses.

1.1 Two distinct exposures: a household's own facilities and its surroundings

A household's own water source and sanitation facility are not the only WASH-related exposure a child faces. Even a child in a household with an improved latrine and piped water can be exposed to pathogens through shared water sources, open defecation by neighbours, flooding, or simple proximity to inadequately managed waste if the surrounding community lacks similar facilities. The distinction between *own-household WASH*, whose protective benefit accrues most directly to the household's own members, and *surrounding-community WASH*, a local environmental condition shaped by the facilities of all households together, is not a modelling convenience: it corresponds to two different exposure pathways. Nor is it a claim that household WASH is a purely private good with no effect beyond its own household. Improved

household sanitation in particular is a textbook source of a local environmental externality, since it reduces the pathogen load that the household itself releases into the shared local environment, which benefits neighbours as well. Recent community-level WASH research (Donohue et al. 2023; Skoufias and Vinha 2026) suggests that household and community WASH need not move together or carry the same association with child growth. For this reason, the thesis constructs and compares both measures rather than treating either as a complete description of a child's WASH exposure: a child's own household classification, and a leave-one-out measure of improved-facility coverage among all *other* households in the same DHS cluster, so that a child's own household is never used to construct its own community exposure.

1.2 Exposure timing in cross-sectional survey designs

A specific measurement limitation affects the cross-sectional, repeated-survey-round observational literature to which the baseline analysis of this thesis belongs. That literature includes DHS-based studies (Rakotomanana et al. 2020), the related national household-survey literature more broadly, such as Addae et al.'s (2024) Ghana MICS-based study, and the baseline of this thesis itself. In these designs, household and community WASH status is recorded once, at the survey date, and applied to every child in that round regardless of the child's age. A four-year-old measured in a 2022 survey experienced their nutritionally critical first two years of life around 2018–2019, under WASH conditions that the survey-date measurement cannot describe, whereas an infant measured in the same round has WASH recorded very close to their own early life. The limitation is specific to designs of this kind. It does not apply to the major WASH-intervention trials, which assign exposure near each child's own birth and follow growth prospectively over a defined window (Section 2.3), nor to panel designs that track WASH and health indicators over successive rounds within the same geographic units (Section 2.2). Age controls and age-interaction tests, which this thesis also estimates, describe how the *current* survey-date association varies with how old a child is today; they do not reconstruct what a particular child's own environment actually was during their own developmental window. The main, baseline analysis of this thesis documents this limitation explicitly and does not claim to resolve it.

A separate, clearly bounded extension responds to one specific dimension of this limitation. It does not ask whether assigning a *historical* instead of a *current* exposure changes an otherwise identical estimate, because the baseline and historical samples differ too much in other respects to isolate that comparison (Section 6.2 sets this out). Instead, the extension asks a self-contained question within its own sample: *is modelled local WASH coverage during a*

child's own early-life window associated with that child's later anthropometric outcomes? To answer it, the extension matches each child's birth-cohort location to a multi-year, modelled local WASH coverage surface for a defined early-life window, using IHME's gridded 2000–2017 drinking-water and sanitation access estimates (Local Burden of Disease WaSH Collaborators 2020). This provides genuine within-cluster, cross-birth-cohort variation that a single survey round cannot offer. The extension improves measurement on one specific dimension, by assigning an exposure closer to each child's own developmental period rather than to the survey date. It does not resolve the exposure-timing problem in general, and it does not reconstruct what any individual household's actual historical WASH facilities were, which no data source available for this thesis can provide. The extension also has its own, separate limitations (undocumented measurement-subsample weighting, DHS/IHME input overlap, residence used as a proxy for historical residence, and confidentiality-driven coordinate displacement). Section 6 states them in full, and they govern how much weight its results can bear.

1.3 Research questions

This thesis asks three related questions. First, is a child's own household's water and sanitation status associated with their anthropometric outcomes (height-for-age, weight-for-age, weight-for-height) after accounting for household- and cluster-level confounding? Second, is the WASH status of the surrounding community, independent of the child's own household, separately associated with these outcomes? Third, within the three countries of the historical sample, is modelled local WASH coverage during a child's specific early-life window associated with these outcomes? All three questions are posed and answered as *associations under a cross-sectional design*, not as estimates of causal effects. This is stated once here and governs how every coefficient in this thesis should be read. The third question is answered only within the historical sample. Its answer is not compared with the answers to the first two questions as if the comparison isolated the effect of exposure timing, for the reasons given in Section 1.2 and developed fully in Section 6.2.

1.4 Scope: four countries, two time horizons

The baseline analysis pools the most recent Demographic and Health Survey (DHS) round available, as of the data-collection date, for four countries: Ethiopia (2024–25), Ghana (2022), Kenya (2022), and Nigeria (2024). These four countries were selected for data availability and recency, not as a representative sample of Sub-Saharan Africa, and no claim in this thesis gen-

eralises findings from these four countries to the region as a whole. The historical extension uses an earlier DHS round for three of the four countries (Ghana 2014, Kenya 2014, and Nigeria 2018), linked to IHME's annual gridded WASH-coverage estimates, which are available through 2017. Ethiopia's 2016 round is retained only as a separately labelled, provisional sensitivity analysis, because the conversion of Ethiopian-calendar interview dates that is required to place Ethiopian children in a specific early-life calendar window could not be fully validated (Section 3.5 and Appendix C). The baseline and historical samples are *not the same children*. They use different exposure definitions and different spatial and temporal constructions, and they are analysed, validated, and reported separately throughout. No result in one sample is used to adjust, select, or validate a result in the other.

1.5 Contribution

This thesis makes two separate contributions in two separate samples, and they should not be read as one combined design. In the *baseline sample*, the contribution is the explicit, side-by-side construction and comparison of own-household and leave-one-out community WASH exposure within one four-country sample and one consistent set of controls, with each exposure estimated under the fixed-effects structure its own construction requires (Sections 4.2–4.3). In the *historical sample*, the contribution is a transparently documented, preliminary attempt to introduce genuine temporal variation in early-life exposure through geospatially matched historical coverage data. This extension examines a single modelled local-coverage exposure assigned to each child's own early-life window (Sections 3.5–3.8). It does not reproduce the household-versus-community comparison: the historical data source does not distinguish a child's own household from its surrounding community, so there is no household/community contrast to draw within the historical sample. A particular combination of four countries is not, by itself, evidence of substantive novelty, and this thesis does not claim it as such. Each of the design choices above is reported with the methodological limitations of its own sample, and these limitations are treated as integral to its interpretation rather than as closing caveats.

1.6 A restrained summary of findings

The main findings are summarised briefly here; Sections 5–6 develop them with full uncertainty. Household-level WASH shows no statistically significant association with height-for-age at the 5% level once cluster fixed effects are used to compare households within the same community (water: $\beta = -0.085$, 95% CI [−0.175, 0.004], $p = .062$, crossing a 10% but not a 5%

threshold; sanitation: β = 0.026, 95% CI [−0.058, 0.109], p = .548). Community-level WASH coverage shows a large, strongly significant unadjusted association for most outcome/exposure pairs, which attenuates to statistical insignificance once socioeconomic controls and region fixed effects are added (water: 0.447 unadjusted vs. 0.033 adjusted, p = .573; sanitation: 0.472 unadjusted vs. 0.008 adjusted, p = .894). The exception is WHZ water, which is already small and non-significant before any adjustment (Section 5.2). This pattern is consistent with, but not conclusive proof of, confounding by local socioeconomic conditions. Within the historical sample, the pooled established-date water–HAZ association is positive and significant (β per 10 percentage points of modelled coverage = 1.03, 95% CI [0.51, 1.54], p < .001). It is, however, sensitive to the weighting convention used (falling to 0.41 unweighted on the same observations) and carries a substantial list of unresolved documentation issues (Section 6; Appendix C). None of these results should be read as evidence either for or against the effectiveness of WASH investment as a policy instrument; they describe statistical associations in specific survey samples, nothing more.

1.7 Design and principal limitations, stated at the outset

This is a cross-sectional, associational design throughout. It is not a panel, difference-in-differences, or instrumental-variables design, and none of those terms is used elsewhere in this thesis to describe a method that is not actually implemented. The principal limitations are unresolved sampling-weight documentation for the measured-anthropometry subsample of the historical extension, DHS/IHME input overlap in two of the three historical-extension countries, the use of current residence as a proxy for historical residence, confidentiality-driven displacement of DHS cluster coordinates, and the small number of countries in both samples. Section 6 states them in full and does not treat them as a closing formality.

1.8 Structure

Section 2 situates this thesis in the literature, distinguishing biological plausibility from whether a facility-level classification shows a detectable association, and causal evidence from conditional association. Section 3 describes the data and variable construction for both samples; Section 4 the empirical strategy; Section 5 the results. Section 6 relates the findings to the literature, explains why the baseline and historical results are not comparable as a test of exposure timing, and states all limitations in full. Section 7 concludes.

2 Literature review and conceptual framework

2.1 From biological plausibility to demonstrated association

The hypothesised pathway from WASH to child growth runs through several distinct, separately evidenced mechanisms, not one undisputed mechanism: inadequate water and sanitation can increase exposure to faecal–oral pathogens, raising diarrhoeal disease, soil-transmitted helminth infection, and sub-clinical environmental enteric dysfunction (EED), each of which may redirect energy and nutrients away from linear growth, though the strength of evidence varies and EED specifically is the least settled (Cumming and Cairncross 2016). Even the strongest recent evidence against basic WASH interventions reducing stunting does not challenge this plausibility: the authors of the three major WASH trials discussed below state explicitly that "the biological plausibility of WASH as public health interventions is not challenged by these findings" (Cumming et al. 2019, consensus message 2), arguing instead that the *specific, basic* interventions tested were insufficient to translate that mechanism into a detectable growth effect. Black et al. (2013) place WASH-related infection among the proximate determinants of child undernutrition in their widely cited *Lancet* framework, alongside diet and care practices, and Victora et al. (2010) document that growth faltering concentrates from birth to roughly 24 months. This review is concerned with a narrower, empirical question: whether a *facility-type classification* — whether a household has an "improved" water source or sanitation facility in the standard WHO/UNICEF JMP sense used throughout this thesis — shows a detectable, policy-relevant statistical association with child growth once the obvious confounders are accounted for, a distinct question from whether WASH-related infection can plausibly affect growth at all.

2.2 Observational evidence and the confounding problem

A substantial observational literature finds WASH variables associated with child anthropometric outcomes in DHS and related survey data. Rakotomanana et al. (2020) link the water and sanitation indicators of the WHO/UNICEF JMP to linear growth in children aged 6–23 months across several East African DHS surveys and find associations in the expected direction. Addae et al. (2024) use Ghana's 2017/18 Multiple Indicator Cluster Survey (MICS), a national household survey comparable to, but distinct from, a DHS round. Among children aged 6–23 months, they find household sanitation and socioeconomic status jointly associated with both stunting and wasting, and unprotected drinking water associated with wasting

but not stunting. Günther and Fink (2010) pool 172 DHS surveys and estimate that improved water and sanitation are associated with lower child mortality, and Spears (2013) documents that cross-country variation in average child height correlates with cross-country variation in open-defecation rates.

The central methodological difficulty in this literature, and one this thesis faces directly, is that household and community WASH status are not randomly assigned: both are strongly correlated with wealth, parental education, and local infrastructure, which independently predict child nutrition through channels unrelated to pathogen exposure. Headey and Palloni (2019) address this directly, constructing a panel of 442 subnational regions across 59 countries from repeated DHS rounds and estimating difference-in-differences regressions that exploit within-region change over time, net of region fixed effects. Their results are mixed and should be read precisely, not as a single attenuation story: improved water access is statistically insignificantly associated with most outcomes (though piped water specifically predicts reduced stunting), and sanitation improvements predict sizeable reductions in diarrhoea and mortality but are *not* associated with stunting or wasting once the within-region design is used. This null sanitation–stunting association, once region-level time-invariant confounding is netted out, is a closer and more precise parallel to this thesis's own null household- and community-sanitation findings (Section 5) than a general "attenuation" story would suggest (Section 6 draws on it this way), and it is not used as evidence that WASH associations "generally" weaken under better controls, which would overstate what this one study establishes.

2.3 Credible causal evidence: what the major trials actually found, and what they did not test

The randomised evidence on WASH and child growth comes primarily from three cluster-randomised controlled trials. They are described here trial by trial and arm by arm, because a loose paraphrase of their design risks two errors: mischaracterising an exposure structure that genuinely differs from that of the cross-sectional literature in Sections 2.2 and 2.5, and attributing a finding from one arm to an intervention other than the one that produced it. WASH-Benefits Bangladesh, WASH-Benefits Kenya, and the Sanitation Hygiene Infant Nutrition Efficacy (SHINE) trial in Zimbabwe each enrolled pregnant women and their children *in utero* and followed them prospectively for 18–24 months. Interventions were assigned near the start of that follow-up window, not at a single arbitrary survey date unrelated to each child's own age. This exposure-timing structure differs materially from that of the DHS-based observational studies discussed in Sections 2.2 and 2.5, and the exposure-timing limitation described

in Section 1.2 for this thesis's own baseline design does not apply to these three trials.

The three trials did not share one common factorial design. WASH-Benefits Bangladesh and Kenya each used a factorial design with **separate arms** for each component (water, sanitation, hygiene, combined-WASH, nutrition with no WASH component, and combined WASH-plus-nutrition), each against a control. SHINE (Zimbabwe) used a simpler 2×2 design with **four arms only** (control, IYCF/nutrition alone, a single combined WASH package with water/sanitation/hygiene bundled rather than separately randomised, and combined IYCF-plus-WASH), so it never tested the three components as independent arms the way the other two trials did. On linear growth, **every individual WASH-related arm in Bangladesh and Kenya showed no statistically detectable effect, and SHINE's single combined WASH arm likewise showed none** — Figure 2 of the consensus paper reflects the design difference directly, referring to "all WASH arms" (plural) for Bangladesh/Kenya and the "WASH arm" (singular) for Zimbabwe (Cumming et al. 2019, 2, 5 and Fig. 2). The nutrition arm, containing no WASH component, showed a separate, modest positive growth effect described by the trial authors as a small gain despite high compliance (Cumming et al. 2019, 6), reported here specifically so this effect is not misread as a WASH finding. The combined WASH-plus-nutrition arm showed **no additional growth benefit over nutrition alone** in any of the three trials (Cumming et al. 2019, 2). On diarrhoea, results were genuinely mixed: a substantial relative reduction in Bangladesh (against a low baseline), no detectable effect in Kenya, and none in Zimbabwe (Cumming et al. 2019, 2–5, Fig. 2).

"No statistically detectable effect" is the precise description used throughout, not "proven to be exactly zero": these are adequately powered trials with high internal validity, but a null result establishes an absence of detected effect at the trial's power, not certainty the true effect is zero, and the authors' own interpretation (discussed next) is explicit that the finding should not be generalised to *every possible* WASH intervention.

The consensus statement states explicitly that the finding does not challenge the biological plausibility of WASH affecting growth (Section 2.1), attributing the null result instead to the limited scope of the specific interventions tested: household-level, "basic" WASH services (JMP ladder sense) that left community-level open defecation and broader water-supply infrastructure unchanged, rather than the "safely managed," networked services the authors argue may be needed for a detectable population-level effect, calling explicitly for research on more comprehensive, "transformative" WASH rather than concluding WASH cannot matter for growth (Cumming et al. 2019). This bears directly on how this thesis's own results should be read: its exposure, household- or community-level "improved" facility status, is likewise a facility-type classification rather than a comprehensive intervention package, so a null or

weak association here is consistent with — though it does not confirm — the trial literature's finding, and equally consistent with confounding masking a real effect, measurement error, or this thesis's own limitations (Section 6); it is not read as evidence that WASH investment in general is ineffective.

A related quasi-experimental literature targets infant mortality, not growth, using a different identification strategy. Geruso and Spears (2015) investigate why Muslim infants in India survive to their first birthday at a substantially higher rate than Hindu infants despite lower average wealth, consumption, and education: neighbourhoods with more Muslim households are *worse* on most observable dimensions, consistent with general socioeconomic disadvantage, except for open defecation, which Hindus practise at a rate roughly 25 points higher nationally (a religious difference unexplained by wealth). Because sanitation and general neighbourhood quality move in *opposite* directions across religious composition, the authors isolate a sanitation-specific channel accounting for a substantial share of the mortality advantage. This thesis cites the working-paper version, for its identification strategy (a source of neighbourhood sanitation variation distinct from general local development) as an example of a design this thesis's own four-country, cross-sectional comparison cannot replicate — not as direct evidence about child growth, which the study does not examine.

2.4 Household versus community exposure

Most DHS-based observational studies reviewed here measure WASH at the household level only; a smaller, more recent strand examines community-level or spatially aggregated WASH exposure as a distinct construct. Donohue, Church, Assaf, and Mayala (2023) construct community-level improved-sanitation coverage (geospatially modelled, not a leave-one-out DHS-census rate) and relate it to stunting in Nigeria and Zambia (both 2018 DHS) via multilevel logistic regression: a mixed, not general, result. In Nigeria, community coverage remained significantly, if modestly, associated with lower stunting odds after full adjustment (adjusted OR = 0.97, 95% CI [0.95, 0.99] per 10-point increase), while household sanitation lost significance in the same model; in Zambia, neither was significant at either stage. This motivates reporting household and community exposures separately, but is not evidence that community coverage generally "carries explanatory relevance beyond" household facilities — that would overstate a result significant in only one of two countries. Skoufias and Vinha (2026), using DHS data from twenty Sub-Saharan African countries, similarly find **no significant externality from community-level water coverage**, but do find community sanitation significantly associated with lower fever/diarrhoea and higher height-for-age,

varying by urban/rural area — motivating considering water and sanitation separately, though this contrasts with this thesis's own fully adjusted community HAZ models, which show no significant association for either measure (Section 5.2). Neither source's result should be read as confirming this thesis's specific coefficients, which come from a different sample, country set, and model. This thesis's leave-one-out community construction follows the same logic as these studies; the author is not aware of a study pairing this exact construction with a household-level comparison in the same four-country sample, though this is not a claim of an exhaustive search — only that this side-by-side design, under the fixed-effects structure each exposure's construction requires, was not found replicated in the sources reviewed here. Balk et al. (2004), on child mortality in West Africa with spatially referenced survey data, provide an earlier precedent for treating DHS-cluster geography as a unit of analysis distinct from the household, applied to mortality rather than anthropometric outcomes.

A specific identification issue arises when community WASH is measured in this way. The leave-one-out rate is a property of the cluster rather than of the individual child. Estimating it with cluster fixed effects, the natural approach for household WASH, would therefore remove nearly all of its substantive variation. The coefficient would then be identified almost entirely by the small mechanical residual that arises because a different household is excluded from each child's own calculation (Section 4 derives this precisely). This thesis therefore uses region, not cluster, fixed effects for the community specification.

2.5 Exposure timing in single-round, cross-sectional survey designs specifically

The limitation introduced in Section 1.2 applies to studies that measure WASH status once, at a single survey date, and relate it to the anthropometric status of children of varying current ages within that round. This fits this thesis's own baseline design and the two anthropometric-outcome studies in Section 2.2, Rakotomanana et al. (2020) and Addae et al. (2024). It does **not** describe Günther and Fink (2010) (mortality/morbidity outcomes, not anthropometric) or Spears (2013) (a cross-country height/open-defecation correlation, not an individual child's outcome at varying current ages), nor the major WASH trials (Section 2.3, which assign exposure prospectively from near birth) or Headey and Palloni's (2019) panel design (successive rounds within the same regions, not a single cross-section). Age controls and age-interaction terms, also estimated for this thesis's baseline sample (Sections 4–5), describe whether the *current* association differs with a child's age today; they do not reconstruct a child's own early-life environment, a different empirical question. This single-round limitation is why the historical extension attempts a different, early-life-window exposure assignment within its own sample

(Section 1.2) — not a reason to expect comparing the two samples to isolate an exposure-timing effect (Section 6.2).

2.6 Geospatial exposure assignment, and a matching-method precedent from outside the WASH literature

Assigning a continuous, gridded exposure surface to a survey respondent's location requires handling two well-documented features of DHS data. First, coordinates are randomly displaced for confidentiality (Burgert et al. 2013; Perez-Heydrich et al. 2013): any extraction method reads pixels around the cluster's *recorded*, displaced coordinates, not its true location, and none recovers the true location or corrects for the (undisclosed, cluster-specific) displacement applied. Second, more than one method exists for reading a value from the surface: a containing-pixel extraction reads the single pixel the point falls in, while a buffer-based extraction averages pixels within a radius (by area or, where available, population weighting). This thesis uses an area-weighted buffer average as its primary method and containing-pixel extraction as a labelled sensitivity (Section 3.6; Section 5.8) — the two are not interchangeable, and the choice is made explicit here rather than left implicit. Johnson, Jacob, and Brown (2013) provide an earlier precedent for linking satellite-derived environmental surfaces (forest cover) to DHS cluster locations for a different child-health outcome. Blom, Ortiz-Bobea, and Hoddinott (2022) match historical, geocoded heat-exposure data to each child's own developmental window across 15 rounds of repeated cross-sectional data from five West African countries, a design structure this thesis's historical extension borrows; their own outcome is child nutritional status, which heat exposure (not WASH) is found to worsen. The study is cited here, and only here, for this matching-method structure — not for any specific numerical finding, nor as evidence about WASH, which it does not examine.

2.7 This thesis's contribution and the gap it addresses

This thesis combines two design elements not found combined elsewhere in this review: household and leave-one-out community WASH compared directly within the same sample and controls (Section 2.4); and, in a separate, bounded, explicitly preliminary extension, an attempt to assign modelled local WASH coverage to each child's own early-life window via geospatial matching (Section 2.6), within the three historical-sample countries. That this specific four-country combination is novel is not, on its own, evidence of a substantive contribution, and this thesis does not rely on country selection for that claim. The claimed methodological con-

tribution is narrower: a direct, consistently controlled comparison of the two exposure levels, and a transparent, fully documented attempt — including its own unresolved issues — at an early-life exposure-matching design that neither Donohue et al. (2023) nor Skoufias and Vinha (2026) implement (both examine only current-round community exposure, not one matched to each child's developmental window); neither source is cited as calling for this extension, which is this thesis's own observation. The gap that remains open, and that this thesis does not claim to close, is the one Headey and Palloni (2019) and the trial literature (Cumming et al. 2019) already identify at a larger scale: credibly separating a WASH-specific effect from the broader package of local development that improved facilities tend to accompany. The historical extension narrows only the exposure-timing dimension of that problem within its own sample; it resolves neither the confounding dimension nor exposure-timing for the baseline sample, since the two samples differ in several other respects at once (Section 6.2).

3 Data and variable construction

This section documents the baseline and historical samples separately throughout. The two samples are built from different DHS rounds, use different exposure constructions, and are validated independently; no claim in this section about one sample should be read as applying to the other. Numerical claims are sourced to the authoritative files listed in Appendix A. This section states what each variable is and how it was constructed; the estimation results themselves are reported in Section 5.

3.1 Baseline sample: unit of analysis, rounds, and pooling

The unit of analysis throughout the baseline sample is the child. The baseline pools the most recent DHS round available, as of data collection, for four countries: Ethiopia (2024–25), Ghana (2022), Kenya (2022), and Nigeria (2024). The full pooled, harmonised child file contains 70,231 children across 4,481 DHS clusters and 114 administrative-region fixed-effect cells (one set of regions per country, pooled). This pooled file is **not** the regression sample for any model. It is the universe before outcome-validity and exposure-classifiability restrictions are applied.

3.2 Baseline anthropometric outcomes and under-five eligibility

Three outcomes are constructed according to standard DHS/WHO anthropometric practice: height-for-age (HAZ), weight-for-age (WAZ), and weight-for-height (WHZ). Each outcome is a standard-deviation z-score relative to the WHO Child Growth Standards reference population (WHO Multicentre Growth Reference Study Group 2006). A child contributes to an outcome only if that outcome's z-score is flagged valid in the DHS recode: the implementing code applies one rule, excluding DHS's own documented special codes (9996–9999) and system-missing values, with no separate measurement-universe filter applied in code. Restriction to the under-five anthropometric-measurement universe is, in practice, a property of which children DHS itself records these fields for, not an additional filtering step this pipeline applies. Because valid measurements are not available for every child in the pooled file, the outcome-specific baseline samples are smaller than the full pooled file: 36,985 children (HAZ), 37,260 (WAZ), and 37,088 (WHZ). Comparing the household and community specifications on identical observations further requires both exposures to be classifiable for every child in the comparison. This gives common baseline samples of 36,040 (HAZ), 36,308 (WAZ), and 36,145 (WHZ), spread across 4,426 DHS clusters. Under-five age is taken uniformly, for all four baseline countries, from the DHS recode's own standard current-age-in-months field (`b19`), the DHS-computed best-available age measure; the baseline pipeline applies no country-varying fallback construction. (The historical extension, described separately in Sections 3.5–3.8, constructs its own `age_m` field with a country-varying reported/calendar-difference rule, for a reason specific to that extension's older DHS rounds. The two are distinct constructions over different samples and must not be conflated.)

3.3 Baseline WASH exposures: household and leave-one-out community

The baseline sample has four exposures: own-household improved/non-improved water and sanitation, and their leave-one-out community counterparts, each following the WHO/UNICEF JMP "improved" facility classification (piped water, protected wells/springs, etc. for water; sewer/septic and improved latrines for sanitation), the same classification independently verified for the historical extension's gridded product (Section 3.5, Appendix A). Household WASH is a household-level binary indicator. **Community WASH is an unweighted leave-one-out proportion among other sampled, classifiable HR households in the same DHS cluster, excluding the child's own household** — not a census of the geographical community, and not an average over other children in the analytical sample. For

each child, the numerator and denominator are built from every *other* surveyed household in the cluster whose status is itself classifiable; households recorded as Unknown or not applicable, and (for Ghana's sanitation core variant) an unclassified biodigester, are excluded from **both** numerator and denominator, not merely treated as unimproved. The resulting rate is a $[0, 1]$ proportion, missing (not imputed) if no other classifiable household exists in the cluster. Improved *shared* sanitation is included in the improved category via a Basic+Limited collapse, following the JMP convention that shared use alone does not make an otherwise-improved facility unimproved; a separate robustness variant treats Ghana's biodigester classification differently. These are the construction choices implemented in `scripts/dhs_harmonization/08_hr_cluster_wash_exposure.py`, documented here as implemented. A child's own classification is never part of its own community exposure, which by design makes community WASH vary mainly *between* clusters rather than within them — Section 4.3 derives how little within-cluster variation this leaves, and why it rules out a cluster-fixed-effects specification for this exposure specifically, unlike for household WASH.

3.4 Baseline covariates, geography, and weights

Household- and child-level covariates include child sex, current age in months (`b19`), maternal age, maternal education (categorical), literacy, household size, birth order, wealth quintile, and urban/rural residence. **The baseline sample is built from country-specific DHS recode files, not IPUMS-DHS**, using the DHS recode's own native field names throughout — different from, and not interchangeable with, the historical extension's IPUMS-harmonized variables (Sections 3.5–3.8). Specifically, it is built from the DHS **Children's Recode (KR)** file (DHS's distinct Births Recode, BR, is not used). The KR file's own weight field is `v005`, **the woman's individual sample weight** (rescaled by the documented 1,000,000 factor), not a household-level weight (`hv005`, which exists only in the Household Recode and is unused here); the KR file attaches each birth to its mother's record, so every child inherits her `v005` directly, and DHS documents no separate per-child weight. This is not `PERWEIGHT`, the different, IPUMS-harmonized field the historical extension uses. The cluster identifier is `v021` (DHS's native PSU code), combined with a country prefix into `country_psu`; this is analogous in purpose to the historical extension's `IDHSPSU` but constructed independently, neither derived from nor equal to it. DHS cluster GPS coordinates are randomly displaced for confidentiality (urban up to 2 km; rural 99% up to 5 km, 1% up to 10 km) (Burgert et al. 2013) — inconsequential for the baseline models, which use no coordinates directly, but directly consequential for the historical

extension's spatial matching (Section 3.5). Baseline models use the `v005`-derived weight, normalised within each estimation sample so each country contributes equal total weight mass regardless of raw sample size — an authorised pooling choice, not a claim about any country's raw size determining its influence (Section 4.5 gives the exact formula).

3.5 Historical extension: rounds, exposure source, and scope

The historical extension links an earlier DHS round for three countries, Ghana (2014), Kenya (2014), and Nigeria (2018), to IHME's annual gridded estimates of local access to improved drinking water (W_IMP) and improved sanitation (S_IMP), available for 2000–2017 (Local Burden of Disease WaSH Collaborators 2020). These product definitions were verified directly from the primary source during this project's development, not inferred from the 0–100 raster scale or file names alone. W_IMP is the sum of piped water (on- or off-premises) and other JMP-improved sources (protected wells and springs, bottled water, rainwater collection, bought water). S_IMP is the sum of sewer/septic sanitation and other JMP-improved facilities (improved and ventilated improved latrines, composting toilets). The source reports each as percent of population with access, on a continuous ~5×5 km surface. This project's own spatial extraction computes an **area average of the raster surface within each child's exposure buffer**, not the source study's own population-weighted administrative aggregate. This distinction is maintained throughout and restated wherever a historical coefficient is reported (Section 5.6; Section 6). Ethiopia's 2016 round is included only as a separately labelled, provisional sensitivity analysis (Appendix C), not as part of the established-date pooled sample. The reason is that the Ethiopian-calendar-to-Gregorian interview-date conversion could not be validated to the same standard as the three established-date rounds. The interview date does not place a child's exposure window — each window's months and exposure value are placed from the birth date alone (Section 3.7; Appendix C) — but it governs eligibility: date validity, the requirement that birth precede the interview, and, for the two postnatal windows, whether the window had completed by the interview date. Appendix C details the three candidate conversions and why one of them, labelled C1, is invalid and excluded from reporting. As stated in Section 1.2, this construction assigns each child a *modelled, area-level* local coverage estimate for a defined early-life window. It is a measurement improvement on the timing dimension specifically, not a record of what that child's own household facilities actually were, which neither this nor any other available data source can provide.

3.6 Historical spatial extraction: buffers, displacement, and coverage

Each child's exposure buffer is centred on its DHS cluster's (displaced) coordinates, using an azimuthal equidistant (AEQD) projection local to that cluster. **The primary buffer radius is 2 km for urban clusters and 5 km for rural clusters** (`scripts/historical_wash/10_spatial_extraction_v3.py`, `RADIUS_M`), stated explicitly here rather than left to be inferred from the DHS displacement distances. The displacement distances are a different matter: they describe the *uncertainty* in a cluster's true location, whereas the buffer radius defines the *area averaged* around its recorded location. A labelled sensitivity (Section 5.8, Appendix C.4) repeats the rural assignment at 10 km, and a separate sensitivity uses containing-pixel assignment instead of the buffer average. AEQD preserves true distance and direction from the cluster's own centre point exactly (relative error $\approx 6\times10^{-12}$); it is not, in general, an equal-area projection, but at these small buffer radii its residual area distortion is negligible, confirmed directly. Within the buffer, the raster's value is averaged using a closed-form disk–polygon overlap computation (each pixel approximated by its corner-projected quadrilateral in the local AEQD plane), not a centroid, nearest-neighbour, or Monte Carlo approximation; this project's own validation found the approximation's own error small at the scales used (median relative pixel-area error $\approx 1.6\times10^{-8}$; maximum mean-exposure difference 0.001 percentage points). Coverage completeness is checked explicitly: a child-year is assigned a valid extraction only when its valid-pixel area meets a documented tolerance (within 10^{-3} of full nominal coverage), and partial- or missing-coverage child-years are excluded rather than assigned a value from incomplete coverage; the reported mean is itself averaged only over valid pixels (confirmed to matter: averaging over the full nominal area instead would shift partial-coverage means by up to 73 percentage points).

3.7 Historical exposure windows and completion rules

Four candidate exposure windows are constructed for each child, applied to the same underlying annual raster series: the calendar year of birth; the nine months before birth (prenatal); the first 12 completed postnatal months (the working main window; Section 6.4 states why); and the first 24 postnatal months. Because IHME's product is annual, a window spanning two calendar years is assigned a weighted average of the relevant years' values (weighted by how many months fall in each), and every touched year must have a verified raster value. The interview-date completion check — excluding a child whose window had not yet elapsed by interview — is **not applied uniformly to all four windows**: it applies only to the two post-

natal windows, where it is substantively necessary. The prenatal window needs no separate check, since it ends at birth and the interview date is always after birth by construction. The birth-year window has no completion check at all, so a child born partway through a year and interviewed before that year ends is still assigned the full annual value — a genuine, disclosed limit on what that window's eligibility rule enforces, not a concealed oversight.

A related but separate limitation concerns the annual resolution of the exposure surface itself: because IHME supplies one value per calendar year, the value used for a window's final touched year can reflect conditions from *after* a child's interview date even when the postnatal completion check is satisfied. This project's construction script flags every such case (a "post-interview-year-used" indicator, tallied and retained) but does not exclude or adjust on this basis — the flag is descriptive, not a filter — so this thesis does not claim every historical exposure value is strictly measured before the outcome it is matched to. The postnatal rule guarantees the *window's months* have elapsed by interview, not that the *annual value* representing the final year contains no post-interview information. Eligibility under any rule still requires a valid interview-date conversion, which is why Ethiopia's three candidates matter even for the birth-year window, whose exposure value does not itself depend on the precise interview date (Appendix C).

3.8 Historical covariates, weights, and the established-date measurement universe

The historical models control for cluster fixed effects, survey-qualified country-by-birth-year terms, child sex, maternal age, maternal education, and marital status, with the child's own completed age in months entered as an additional linear control. The country-by-birth-year terms are each country's own birth-year effects relative to that country's own earliest observed birth year, rather than one reference year shared across all three countries. Section 4.6 describes this convention and reports a dedicated diagnostic establishing that, once the cluster fixed effect and other covariates are accounted for, this specific choice does not itself change the fitted exposure coefficient, though it does change the raw design matrix. **The age control is constructed differently from the baseline's age variable, not identically.** The historical extension uses its own `age_m` field: reported completed months where delivered directly by the source recode (Ethiopia, Nigeria), and a documented calendar-month-difference construction otherwise (Ghana, Kenya). The baseline, by contrast, uses DHS's own standard `b19` field uniformly for all four countries, with no country-varying construction (Section 3.2). The two fields happen to serve an analogous purpose (a linear age-in-months control) but are built

by different code over different samples, and neither is a copy of the other's rule. Household wealth quintile is **excluded from the historical extension's main specification**, for a documented reason that does not rest on fixed effects: current wealth may be a mediator or post-exposure outcome of the historical WASH exposure, and is correlated with the exposure itself, so including it risks conditioning on a variable downstream of the exposure of interest. **This is a genuine trade-off, not a one-sided argument for exclusion.** The competing concern is that current wealth, measured at the survey date, could instead (or also) proxy a household's *prior* socioeconomic circumstances during the child's early-life window — circumstances that could themselves confound the WASH–nutrition association, in which case omitting wealth leaves that confounding unaddressed rather than avoiding over-adjustment. Correlation between wealth and the exposure is not, on its own, a sufficient reason to omit a potential confounder; it is a reason to treat wealth's role as a substantive assumption requiring disclosure, not an automatic exclusion rule. Wealth is therefore included only as a labelled sensitivity check (Section 5.8), where the pooled water–HAZ coefficient is essentially unchanged (1.01 vs. 1.03) — showing the reported coefficient is not sensitive to this adjustment, without by itself adjudicating between the mediator and confounding readings. This reason is different from, and more defensible than, the claim that "cluster fixed effects absorb wealth": that claim is false in general, since cluster fixed effects absorb only what is constant *within* a cluster, and wealth varies across households within the same cluster just as WASH status does, confirmed directly in this thesis's own baseline cluster-fixed-effects model, where wealth quintile *is* included and estimated as a standard covariate (Section 4.2).

Weighting uses PERWEIGHT, normalised so that each of the three established-date countries contributes an equal total observation weight within each model's own final estimation sample (not population-size weighting). This is the same authorised convention as the baseline, carried over for comparability, and it is reported alongside its unweighted counterpart for the headline cell (Section 5.6). The documented sampling weight for measured anthropometry specifically, as distinct from the general household/individual sampling weight, could not be located in official DHS documentation despite repeated attempts at alternative official sources (Appendix A records the exact sources tried and their outcomes). This project's working inference, stated explicitly wherever historical coefficients are reported, is that within-survey PERWEIGHT normalisation removes a constant measurement-subsample selection factor under an assumption of random household-level selection into the anthropometry subsample. It cannot and does not correct for any differential non-response among eligible children who were not measured. Historical results are therefore reported as **weighted conditional associations among eligible measured children**, not as population-representative estimates for any

defined population, and this phrasing is used consistently rather than varied across sections.

3.9 A documented measurement-universe discrepancy: Nigeria 2018

Three distinct child populations arise for Nigeria 2018, and this thesis does not force them to a common count or treat the gap as resolved. The official 2018 NDHS report counts 12,806 (unweighted) children under five eligible and actually measured (men's-questionnaire subsample households, roughly a third of all households). The IPUMS-DHS birth-history extract used here contains 33,924 birth-history records (children born within approximately five years of the survey), of which 11,704 are non-missing for anthropometry. The final historical-extension estimation sample (water, HAZ, main post-12-month window) retains 6,912 after the window-completion, spatial-coverage, and singleton-cluster exclusions (Section 4.5). These differ by construction: the official count is eligible children in subsample households; the extract is birth-history children of interviewed women, which can omit children whose mothers are not in the sampled frame (raised in this project's documentation, not independently verified); the estimation sample further requires complete windows and spatial coverage. The gap between the official count and the extract (12,806 vs. 11,704) is disclosed as an open, unresolved question, not treated as resolved by any estimation choice here.

4 Empirical strategy

Throughout, i indexes a child, h the child's household, c the child's DHS cluster, r the child's administrative region (each cluster belongs to exactly one region), and s a survey round (country). All specifications below are estimated and reported in this thesis as **associational**, not causal; none should be read, described, or cited elsewhere in this thesis as identifying a causal effect. None of the designs below is a panel design, a difference-in-differences design, or an instrumental-variables design. These terms are not used anywhere in this thesis to describe a specification that is not what is actually estimated.

4.1 Baseline: household WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta_W \,\text{Water}_{ih} + \beta_S \,\text{Sanitation}_{ih} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

Water_{ih} and Sanitation_{ih} are the child's own household water and sanitation status, **entered together in the same model** (every household-WASH specification in this thesis estimates β_W

and β_S jointly, never one at a time). X_i is the covariate set of Section 3.4, and δ_r a fixed effect for each of the 114 pooled administrative regions, so β_W and β_S are identified from **within-region** variation only, net of controls and of the other WASH measure. A region can contain many DHS clusters, so δ_r still allows clusters within it to differ in local infrastructure, environment, or health-service access — characteristics plausibly correlated with both household WASH and child nutrition that this specification does not absorb separately, motivating the cluster-fixed-effects specification below, which narrows the comparison to households within the same cluster.

4.2 Baseline: household WASH, DHS cluster fixed effects (the main household specification)

$$Y_{ihc} = \alpha_c + \beta_W \,\text{Water}_{ih} + \beta_S \,\text{Sanitation}_{ih} + X_i'\gamma + \varepsilon_{ihc}$$

As in Section 4.1, Water_{ih} and Sanitation_{ih} are entered together, mutually adjusted; α_c is a separate fixed effect for each DHS cluster, so β_W and β_S are identified from differences in household WASH status *among households within the same cluster*, conditional on the covariates and each other. α_c absorbs anything every household in a cluster shares (local infrastructure, environment, market access) but not differences *between* households within it — neighbouring households can still differ in wealth, education, or unobserved behaviour connected to both WASH status and growth, which is why this specification remains associational. One mechanical consequence: since every DHS cluster is entirely urban or entirely rural by sampling design, urban/rural residence is automatically absorbed by α_c and cannot be estimated separately. **Wealth quintile is included in X_i and is estimated, not omitted**: cluster fixed effects absorb only what is constant within a cluster, and wealth varies across households within one just as WASH status does (31.6% of clusters show within-cluster variation in own-water status, 50.6% in own-sanitation, the variation that identifies β_W/β_S; an identical check confirms comparable within-cluster variation in wealth), so there is no mechanism by which cluster fixed effects would remove it, and none was observed.

4.3 Baseline: community WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta_W \bar{C}^{water}_{-h,c} + \beta_S \bar{C}^{san}_{-h,c} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

$\bar{C}^{water}_{-h,c}$ and $\bar{C}^{san}_{-h,c}$ are the leave-one-out community water and sanitation coverage rates defined in Section 3.3. They are again **entered together and mutually adjusted**, including in the "minimally adjusted" stage of Table 5.2, which still adjusts each community measure for

the other even though it adjusts for nothing else. These exposures are, by construction, properties of the cluster rather than of the individual child. Here β_W and β_S are therefore identified mainly by comparing *clusters* with different community coverage within the same administrative region, net of the same controls. As with the household region-fixed-effects model, unobserved differences between clusters inside the same region are not removed. This is an associational estimate of a contextual exposure, not an estimate of a causal spillover effect.

Why cluster fixed effects are not used for this exposure. Community WASH could, in principle, also be estimated with cluster fixed effects, but the resulting estimate would not be informative, for a specific, verifiable reason: the leave-one-out rate is itself a cluster-level quantity, and within a cluster, each child's value differs from every other's only because a different single household is excluded from each calculation — a mechanical artefact. Measured directly, the within-cluster standard deviation of this rate is approximately 0.007 (water) and 0.011 (sanitation), against an overall standard deviation of approximately 0.35 for both, so a cluster fixed effect would absorb nearly all genuine between-cluster variation, identifying the coefficient almost entirely from this small mechanical residual rather than real differences in community environments. The historical extension (Section 4.6) differs fundamentally: a real time dimension (birth-cohort variation within cluster) gives its cluster fixed effects genuine, non-mechanical variation to work with. Cluster fixed effects are therefore appropriate for household and historical WASH but not current-round community WASH — a design consequence of the leave-one-out construction, not an inconsistency.

4.4 Baseline: household and community WASH jointly, and current-age heterogeneity

$$Y_{ihcr} = \alpha + \beta_{HW}\,\text{Water}_{ih} + \beta_{HS}\,\text{Sanitation}_{ih} + \beta_{CW}\,\bar{C}^{water}_{-h,c} + \beta_{CS}\,\bar{C}^{san}_{-h,c} + X_i'\gamma + \delta_r + \varepsilon_{ihcr}$$

All four WASH terms (household water, household sanitation, community water, community sanitation) are entered **together, in one model**, and each coefficient is conditional on the other three. β_{HW}/β_{HS} are the household associations conditional on surrounding community coverage and on each other; β_{CW}/β_{CS} are the community associations conditional on the household's own status and on each other. Because household and community water measures are fairly closely correlated, this specification is reported as a secondary decomposition, not a preferred model.

$$Y_i = \alpha + \beta_W \,\text{Water}_i + \beta_S \,\text{Sanitation}_i + \sum_{a \neq \text{ref}} \theta_a \,\text{AgeBand}_{ia}$$
$$+ \sum_{a \neq \text{ref}} \lambda_a^W \,(\text{Water}_i \times \text{AgeBand}_{ia}) + \sum_{a \neq \text{ref}} \lambda_a^S \,(\text{Sanitation}_i \times \text{AgeBand}_{ia})$$
$$+ X_i' \gamma + \text{FE} + \varepsilon_i$$

Water and sanitation, and both of their full interaction blocks, are entered **together in one model**, exactly as in every other household specification in this thesis. AgeBand_{ia} are six current-age bands in completed months (0–5, 6–11, 12–23, 24–35, 36–47, 48–59). They replace the continuous age control used elsewhere, with **12–23 months as the reference band**, fixed before any heterogeneity result was examined. Each of the other five bands therefore contributes one θ_a main-effect dummy and one interaction term per exposure (λ_a^W, λ_a^S), ten interaction terms in total. Three joint Wald tests are reported per outcome and fixed-effect structure: a water-specific test ($H_0 : \lambda_a^W = 0 \;\forall a$, 5 restrictions), a sanitation-specific test (5 restrictions), and a combined test of all ten interaction terms jointly (10 restrictions). Each is referred to a χ^2 distribution with its stated degrees of freedom, not an F distribution. The $\lambda_a^W / \lambda_a^S$ terms test whether the association between *survey-date* WASH and nutritional outcomes differs across children's *current* ages. This is a heterogeneity test on the existing cross-sectional design (Section 1.2). It is not a reconstruction of what WASH conditions an older child actually experienced during their own infancy, and it is kept conceptually and terminologically distinct from the historical extension's exposure-timing design throughout this thesis.

4.5 Inference, weighting, and the pooled estimand

Weighting. For a model with an estimation sample of size N pooled across M countries, every country is given equal total weight in the pooled regression. Both pipelines implement this with each child's weight computed as

$$w_i = \text{raw}_i \Big/ \sum_{j \in \text{country}(i)} \text{raw}_j$$

so that each country's weights sum to exactly 1. Here raw_i is the sample's own native sampling-weight field: `v005`-derived for the baseline (Section 3.4) and `PERWEIGHT` for the historical extension (Section 3.8), never the same field for both, computed within that model's own final estimation sample after all exclusions actually applied to it (including the historical pipeline's singleton-cluster removal, see below; the baseline pipeline applies none). The historical pipeline's own implementation (`scripts/historical_wash/16_estimate_preliminary.py`) additionally multiplies this weight by the con-

stant N/M, so each country's weights sum to N/M rather than 1; the baseline pipeline's implementation (`build_temporary_pooled_weight()` in `scripts/dhs_harmonization/11_main_regressions.py`) does not. This difference has **no effect on any reported coefficient, standard error, confidence interval, or p-value**, since N/M is the same constant for every observation in a given model and WLS with its cluster-robust sandwich variance is invariant to rescaling every weight by one common constant (confirmed directly by refitting under both conventions: identical coefficients and SEs to machine precision). The two pipelines' conventions are therefore substantively identical (equal total country weight), differing only in a reporting-level normalisation constant. (M denotes the number of countries, deliberately a different symbol from K, the regressor count used below.) **This equal-country-total weighting is an authorised, disclosed working choice, not a correction for a sampling design, and it does not, by itself, define the pooled coefficient as any particular simple average.**

The actual estimand is the single common slope β of one pooled weighted regression, run jointly across all countries' observations at once — not the result of a two-step procedure that estimates each country's own slope separately and averages them. Equal country weight *totals* do **not** make the pooled coefficient the arithmetic mean of the country-specific coefficients: by the Frisch–Waugh–Lovell logic underlying any partialled-out regression coefficient, the pooled β is a weighted average of each country's own residualized-exposure contribution, where the weight each country effectively receives depends jointly on its observation weights **and** on the weighted sum of its *squared* residual exposure after the fixed effects and controls are partialled out. A country with little residual within-cluster exposure variation therefore contributes comparatively little statistical information to the slope, even at an equal weight total. This thesis's own historical pooled water–HAZ estimate illustrates the point as a documented empirical pattern, not a formal decomposition: the simple unweighted average of the three established-date countries' own coefficients is 0.58, while the actual pooled coefficient is 1.03 (Section 5.6), a pattern consistent with Kenya's residual exposure variation after controls (0.20 percentage points) being roughly a third of Nigeria's (0.65) and under half of Ghana's (0.46), so that Kenya's imprecise, negative country-specific estimate would be expected to receive comparatively little weight despite its equal weight total. For the primary established-date models, each pooled coefficient is reported alongside its country-specific components (Tables 5.5 and 5.6) and the relevant residual exposure variation (this section; Section 4.6's range across countries and products), so a reader can see which countries drive the headline pooled result rather than inferring equal influence from the weighting description.

Estimator and inference, stated separately for each pipeline because they are not

identical. Household- and historical-model cluster fixed effects are both implemented as weighted within-cluster demeaning, exactly equivalent to weighted least squares with a full set of cluster dummy variables, using the cluster-robust CRV1 sandwich estimator with the small-sample correction $c = \frac{G}{G-1} \times \frac{N-1}{N-K}$ (G = clusters, K = non-fixed-effect regressors retained — the project's adjudicated convention when the fixed-effect and clustering groupings coincide). **The two pipelines differ, however, in the reference distribution used for confidence intervals and p-values, and this thesis describes each as implemented.** The historical pipeline (`scripts/historical_wash/16_estimate_preliminary.py`) uses the Student t-distribution with $G - 1$ degrees of freedom. The baseline pipeline (the `statsmodels` (Seabold and Perktold 2010) calls in `scripts/dhs_harmonization/14_household_community_wash_models.py`) does **not**: its executed `run_log.txt` records `use_t: False`, and independent recomputation confirms the stored p-values match a **normal (z)** reference distribution, not $t(G-1)$, to more than ten decimal places (the household water–HAZ cell's two candidate values, .0617391913770433 vs. .0618052, both round to the displayed .062, which is why a rounded-table check alone would miss the difference). The region-fixed-effects models (Sections 4.1, 4.3, 4.4) are fitted as weighted least squares with explicit region dummies clustered on `country_psu`, counting the region dummies and the constant in K, and likewise use the normal reference distribution. This thesis's own primary baseline household cluster-FE p-values (water HAZ .062, sanitation HAZ .548) are therefore normal-reference values; the alternative, more conservative convention that additionally penalises degrees of freedom for the absorbed cluster dummies gives wider sensitivity p-values (.080, .574), reported as such wherever cited, never silently substituted. This difference between pipelines is a documented implementation fact, not changed here merely to make the two descriptions match.

Singleton-cluster treatment, stated separately for each pipeline. Before estimation, the historical pipeline explicitly drops clusters that contribute only one observation to a given model's final estimation sample (the `sizes >= 2` filter in `16_estimate_preliminary.py`) and reports the number dropped with every model. A singleton cluster carries no within-cluster information and cannot contribute to identifying a cluster-fixed-effects coefficient. **The baseline pipeline's household- and community-model code (`14_household_community_wash_models.py`) contains no corresponding singleton-exclusion step.** This thesis therefore does not assert that baseline singletons are dropped, and it does not report a baseline singleton count, because no such step exists in the code that produced the baseline tables in Section 5. This is disclosed as a description gap,

not assumed to be numerically inconsequential. Sample size, cluster count, the finite-sample correction c above, and the model-specific country-weight normalisation (Section 4.5) could all be affected by whether singleton clusters are present, and this thesis does not claim to have quantified that effect for the baseline models.

Rank and identification are computational properties, not causal ones. Every reported model's design matrix is checked for full column rank after the fixed-effects transformation. The historical pipeline drops any fully absorbed column and records it by name; the baseline pipeline stops if the design is rank-deficient, and its only pre-specified exclusion (`residence` under cluster fixed effects) is stated in Section 4.2. Neither pipeline silently reports an unstable coefficient. This establishes that the stated regression equation is computationally well defined and that its coefficient is the one the equation describes. It does not establish that the equation identifies a causal parameter, and this thesis does not conflate the two.

4.6 Historical extension: cluster fixed effects with cohort controls

$$y_{ic} = \alpha_c + \beta\, E_{ic}(w,p) + \lambda\, \mathrm{age}_{ic} + \gamma' X_{ic} + \delta_{s,b_{ic}} + \varepsilon_{ic}$$

$E_{ic}(w,p)$ is the annual-weighted IHME mean (product p, window w) assigned to child i in cluster c, on the 0–100 percentage-point scale (Sections 3.5–3.7); β is reported per 1 and per 10 percentage points. α_c is the cluster fixed effect. $\delta_{s,b}$ is a survey-specific (country-specific) birth-year term: each country's own set of birth-year fixed effects, estimated relative to that country's own earliest observed birth year rather than to one pooled reference year across all three countries. An early, incorrect independent-reconstruction attempt (Appendix B §B.2) both used a single pooled reference cell for the birth-year terms *and* omitted the child's own age-in-months control. Together, these two changes produced a materially different coefficient (1.385 vs. the correct 1.026). A dedicated diagnostic (`scripts/historical_wash/25_diagnose_birthyear_coding_and_age_control.py`) isolates the two changes rather than leaving them conflated. Its comparisons never use the exposure variable itself to construct the test: residualizing a candidate control set against the exposure before checking whether it affects the exposure's own coefficient would be circular, and this diagnostic does not do that.

A single pooled reference year does mechanically change the raw design matrix (Nigeria gains an extra year dummy relative to the country-local scheme), but this extra column adds no genuine dimension: it sums exactly to Nigeria's own country indicator, which the cluster fixed effect already fully absorbs. Consequently the two schemes' nuisance specifi-

cations span equal subspaces both before and after residualizing against the shared controls, so the exposure variable's own residualized values — and hence its fitted coefficient — are identical between schemes by the Frisch–Waugh–Lovell theorem. This is confirmed directly, not merely inferred: an independent partial-regression coefficient and the fitted models themselves agree between schemes to floating-point precision (absolute difference 6.7×10^{-16}). The age-in-months control, by contrast, accounts for the full 1.385-versus-1.026 gap on this reconstruction ($\beta = 1.0255531$ with it, matching the stored 1.025553; $\beta = 1.3852855$ without it). This diagnostic is a reconstruction of the described design, not a byte-for-byte rerun of its original, unpreserved code, so this thesis states it as evidence the age-in-months omission explains the historical discrepancy on this reconstruction, not as certainty about what that specific, no-longer-existing script did. The full step-by-step derivation is in Appendix B §B.2. X_{ic} includes child sex, maternal age, maternal education, and marital status. Wealth is excluded from the main specification for the reason given in Section 3.8, not because cluster fixed effects absorb it.

Why cluster fixed effects are appropriate here but not for current-round community WASH (Section 4.3). The historical exposure $E_{ic}(w, p)$ varies not only across clusters but also, genuinely, across birth cohorts *within* the same cluster. A child born in 2014 and a child born in 2017 in the same cluster can have experienced different modelled local WASH coverage during their own respective early-life windows. This is real temporal variation, not a mechanical construction artefact of the kind that rules out cluster fixed effects for the current-round leave-one-out community measure. The cluster-fixed-effects specification is therefore appropriate for the historical exposure, conditional on there being enough genuine within-cluster, cross-cohort variation for the fixed effect not to absorb nearly all of it. That variation is quantified, per model, as the residual exposure standard deviation after all controls (Section 5.6), which ranges from 0.20 to 0.88 percentage points depending on country and product. This range is modest relative to the 10-percentage-point scale conventionally used to report the coefficient (Section 5.6 states the ratio explicitly), and this thesis does not interpret a 10-point coefficient as describing an exposure contrast that is typically observed within a cluster.

Weighting follows the same equal-country-total formula as Section 4.5, with $M = 3$ countries for the established-date pooled models (Ghana, Kenya, Nigeria). Inference uses the Student $t(G-1)$ reference distribution that the historical pipeline actually implements (Section 4.5 states precisely how this differs from the baseline pipeline's normal-reference inference). Model-specific samples are constructed independently for every window and product combination. Singleton clusters are dropped and counted by the historical pipeline's own explicit filtering step; this is a property of the historical pipeline's code specifically, not a claim that

the baseline pipeline does the same (Section 4.5).

5 Results

Every table in this section is taken directly from the authoritative, validated output files listed in Appendix A; no coefficient was computed for this chapter. Coefficients are reported with 95% confidence intervals, except in Table 5.4, which reports coefficients and p-values only. P-values are shown to three decimal places when the underlying value is at least .0005 and as "<.001" when it is smaller. Both forms are display precision only; the full-precision value of every cell is stored in the authoritative output files. Where a model has two inference conventions, both are labelled explicitly (Section 4.5), and the prose uses the primary convention unless the sensitivity convention is named.

5.1 Household WASH

Table 5.1 — Household WASH: region-FE and cluster-FE results.

Outcome	Water β [95% CI], *p*	Sanitation β [95% CI], *p*	N	Clusters
Region FE				
HAZ	−0.035 [−0.110, 0.040], .357	0.022 [−0.045, 0.089], .522	36,040	4,426
WAZ	0.021 [−0.038, 0.081], .487	0.045 [−0.010, 0.100], .109	36,308	4,426
WHZ	0.050 [−0.015, 0.116], .130	0.046 [−0.011, 0.103], .116	36,145	4,426
Cluster FE (main household specification; PRIMARY inference convention)				
HAZ	−0.085 [−0.175, 0.004], .062	0.026 [−0.058, 0.109], .548	36,040	4,426
WAZ	−0.029 [−0.100, 0.043], .435	0.052 [−0.015, 0.120], .130	36,308	4,426
WHZ	0.012 [−0.062, 0.085], .758	0.053 [−0.014, 0.119], .122	36,145	4,426

Household water and sanitation are entered together in every model in Table 5.1 (Sections 4.1, 4.2). No household coefficient reaches significance **at the 5% level** in the cluster-fixed-effects specification, this thesis's main household result and the one least exposed to cluster-level confounding; water–HAZ (p = .062) does cross the 10% threshold, stated explicitly rather than left implicit. Non-significance is read as inconclusive, not as proof no association exists — the water–HAZ interval, for example, spans −0.175 to +0.004. Under the more conservative FULL_DUMMY_DF sensitivity (Section 4.5), the cluster-FE p-values are .080 (water–HAZ) and .574 (sanitation–HAZ), the same substantive conclusion. These use the normal reference distribution the baseline pipeline's code actually computes (Section 4.5), not

the Student $t(G-1)$ distribution used for the historical extension (Section 4.6).

5.2 Community WASH: the adjustment path

Table 5.2 — Community WASH: the four-stage adjustment path (HAZ).

Stage	Water β [95% CI], *p*	Sanitation β [95% CI], *p*
Minimally adjusted	0.447 [0.311, 0.584], <.001	0.472 [0.356, 0.587], <.001
+ child controls	0.395 [0.263, 0.527], <.001	0.452 [0.340, 0.565], <.001
+ socioeconomic status	0.127 [0.010, 0.244], .034	0.008 [−0.117, 0.133], .900
+ SES + region FE (fully adjusted)	0.033 [−0.083, 0.150], .573	0.008 [−0.113, 0.129], .894

Note. HAZ; N = 36,985 and 4,434 clusters in all four stages. This is the outcome-valid sample. It is slightly larger than the 36,040-child, 4,426-cluster common sample used in Table 5.1 and Table 5.3, because this table's community-only specification does not also require household WASH to be classifiable for every child. The attenuation pattern described below should therefore not be compared cell for cell with Table 5.1's household coefficients as if the two samples were identical. Table 5.3's joint model, estimated on the common sample, is this thesis's like-for-like comparison of household and community exposure.

The community water and sanitation coefficients, which are entered together at every stage (Section 4.3), are large and highly significant before adjustment. They attenuate sharply as socioeconomic status and region fixed effects are added, and neither is significant at the fully adjusted stage. **This attenuation pattern is consistent with confounding by local socioeconomic conditions, but it does not, on its own, establish confounding as the sole explanation.** The same pattern could also arise, for example, from a change in measurement structure as additional controls absorb variance, or from other specification features that happen to coincide with the added controls. The same four-stage pattern holds for WAZ and WHZ, with one difference that should be stated precisely. The WHZ water association is already small and statistically insignificant before adjustment (0.033, $p = .545$) and remains so throughout, whereas the HAZ and WAZ water associations start large and attenuate to insignificance. Across all three outcomes, the fully adjusted specifications (the community models here and the household models in Table 5.1) do not show a consistent association between improved WASH and child anthropometric outcomes, at either the household or the community level. This conclusion is read as "this sample does not provide clear evidence of an association once the obvious confounders are controlled", not as "WASH has no true relationship with child growth", which is a materially stronger claim than this design can support.

5.3 Joint household/community model

Table 5.3 — Joint household/community model: all four WASH terms entered together.

Exposure	β [95% CI]	*p*
HAZ		
Household water	−0.085 [−0.167, −0.003]	**.043**
Household sanitation	0.024 [−0.047, 0.096]	.507
Community water	0.120 [−0.008, 0.248]	.066
Community sanitation	−0.018 [−0.148, 0.112]	.784
WAZ		
Household water	−0.015 [−0.084, 0.055]	.677
Household sanitation	0.039 [−0.020, 0.098]	.191
Community water	0.081 [−0.021, 0.184]	.121
Community sanitation	0.022 [−0.092, 0.136]	.706
WHZ		
Household water	0.039 [−0.038, 0.117]	.318
Household sanitation	0.036 [−0.025, 0.096]	.244
Community water	0.019 [−0.099, 0.137]	.750
Community sanitation	0.045 [−0.070, 0.160]	.446

Note. N = 36,040 for the HAZ model, 36,308 for WAZ and 36,145 for WHZ, with 4,426 clusters in each. Each outcome uses its own common sample, so the sample is not literally identical across outcomes.

One cell in this table reaches significance at the 5% level: household water for HAZ, conditional on community water and the other two WASH terms (β = −0.085, p = .043), and its sign is unexpectedly negative. Household and community water measures are correlated, and this is one significant cell among the twelve estimated in this table alone. The result is therefore reported as a single finding rather than treated as a central conclusion of this thesis, and it is not interpreted as evidence that improved household water is harmful.

5.4 Specification sensitivity: Nigeria community sanitation

Table 5.4 — Nigeria community sanitation–HAZ: specification-sensitivity results.

Specification	Sanitation–HAZ β	*p*
No controls	0.274	.004
+ child controls	0.296	.001
+ socioeconomic controls	−0.457	<.001
+ region FE	−0.184	.035
Household cluster FE (own-household sanitation)	−0.007	.919

The Nigeria community sanitation–HAZ coefficient changes sign and grows in absolute size as controls are added sequentially, from 0.274 with no controls to −0.457 once socioeconomic controls are included. A block-by-block audit finds that removing wealth quintile alone from the full control set moves the coefficient from −0.457 back to −0.196, the largest movement produced by any single control block. This demonstrates sensitivity to specification; it is not proof that wealth is the specific mechanism behind the reversal. The corresponding household-level (cluster-fixed-effects) coefficient for Nigeria is close to zero and not significant. This instability is reported as evidence that the community-level, region-fixed-effects estimate for Nigeria sanitation is **not stable across specifications**, and it should not be read as evidence that improved sanitation harms child growth in Nigeria.

5.5 Current-age heterogeneity

In the heterogeneity models, six current-age bands (0 to 5, 6 to 11, 12 to 23, 24 to 35, 36 to 47, 48 to 59 completed months) replace the continuous age control, 12–23 months as reference (Section 4.4). Of twelve exposure-specific joint interaction tests (two exposures × three outcomes × two fixed-effects structures, each a χ^2 test of five restrictions), **eleven of twelve reject the null of no heterogeneity** at the 5% level (the exception: the water interaction for WHZ under region fixed effects, $p = .300$); the combined ten-term tests (χ^2, 10 df) reject in all six combinations (largest $p = .019$). The household-WASH association with current outcomes therefore varies with current age in most specifications tested. As Section 4.4 states, this is secondary evidence about the *current* survey-date association only — it compares children of different current ages at one survey date, and does not trace growth over time or reconstruct any child's actual early-life exposure, a question addressed only by the historical extension below, under its own separate limitations.

5.6 Historical extension: primary established-date models

All cells below come from the primary weighted models with the post-12-month window, estimated on the established-date sample (Ghana 2014, Kenya 2014, Nigeria 2018). Coefficients are per 10 percentage points of IHME-modelled local coverage. The exposure is an area average of the modelled coverage surface within each child's buffer; it is neither the source study's population-weighted aggregate nor a record of the child's own household facilities (Section 3.5). The estimates are weighted conditional associations among eligible measured children, not population-representative estimates (Section 3.8). Their confidence intervals and

p-values use the Student $t(G-1)$ reference distribution (Section 4.5). Every cell carries the same methodological qualifications, stated in Section 3.8 and discussed fully in Section 6, and none is promoted above the others for being significant.

Table 5.5 — Historical extension: primary established-date water (W_IMP) estimates.

Water (W_IMP): coefficients per 10 percentage points

Sample	β [95% CI]	*p*
HAZ		
Ghana 2014	1.65 [0.34, 2.96]	.014
Kenya 2014	−0.61 [−2.07, 0.86]	.417
Nigeria 2018	0.70 [0.15, 1.25]	.013
Pooled established	**1.03 [0.51, 1.54]**	**<.001**
WAZ		
Ghana 2014	1.06 [−0.16, 2.28]	.088
Kenya 2014	−0.22 [−1.30, 0.86]	.690
Nigeria 2018	0.45 [0.00, 0.90]	.050
Pooled established	0.65 [0.20, 1.10]	.005
WHZ		
Ghana 2014	−0.00 [−1.16, 1.16]	.997
Kenya 2014	−0.16 [−1.29, 0.96]	.776
Nigeria 2018	−0.05 [−0.48, 0.38]	.809
Pooled established	−0.08 [−0.50, 0.35]	.717

Table 5.6 — Historical extension: primary established-date sanitation (S_IMP) estimates.

Sanitation (S_IMP): coefficients per 10 percentage points

Sample	β [95% CI]	*p*
HAZ		
Ghana 2014	−2.48 [−4.91, −0.05]	.046
Kenya 2014	−0.21 [−0.53, 0.11]	.190
Nigeria 2018	0.20 [−0.34, 0.75]	.467
Pooled established	−0.10 [−0.37, 0.17]	.483
WAZ		
Ghana 2014	−0.08 [−2.04, 1.88]	.935
Kenya 2014	−0.15 [−0.38, 0.09]	.223
Nigeria 2018	0.02 [−0.44, 0.48]	.937
Pooled established	−0.07 [−0.29, 0.14]	.511
WHZ		
Ghana 2014	1.64 [−0.27, 3.56]	.092
Kenya 2014	0.00 [−0.22, 0.22]	.980
Nigeria 2018	−0.17 [−0.59, 0.24]	.407
Pooled established	−0.03 [−0.24, 0.17]	.742

Table 5.7 — Historical extension: primary established-date sample sizes (N children / G clusters).

Outcome	Ghana 2014	Kenya 2014	Nigeria 2018	Pooled established
HAZ	N1,940 / G353	N14,166 / G1,498	N6,912 / G1,263	N23,018 / G3,114
WAZ	N1,940 / G353	N14,166 / G1,498	N6,951 / G1,264	N23,057 / G3,115
WHZ	N1,940 / G353	N14,166 / G1,498	N6,942 / G1,264	N23,048 / G3,115

Note. The sample sizes are identical for the water and sanitation estimates in Table 5.5 and Table 5.6, because both exposures are estimated on the same anthropometric sample for each outcome and country. They are therefore shown once here rather than repeated in every coefficient cell.

The pooled established-date water–HAZ coefficient (1.03, $p < .001$) is this section's selected headline, the cell this thesis discusses in most depth. It is **not** the only nominally significant cell among the 24 shown above. At conventional $p < .05$, **six** of the 24 cells reach significance: pooled water–HAZ, pooled water–WAZ, Ghana water–HAZ, Nigeria water–HAZ, Nigeria water–WAZ ($p = .0498$) and Ghana sanitation–HAZ, the last in an unexpected negative direction. The displayed, rounded $p = .050$ for Nigeria water–WAZ in Table 5.5 should not be read as lying exactly at the threshold, because the underlying value is below .05. The pooled water–HAZ cell was chosen as the headline because it is the pooled, established-date estimate for this thesis's primary outcome and primary window, a choice made before any coefficient was examined (Section 6.4), not because it is the only significant result. A reader comparing the tables with this text should therefore see six significant cells, not one.

All 24 primary cells in Table 5.5 and Table 5.6, including this headline, were independently reconstructed and re-estimated with a different estimation library on a separately rebuilt sample, matching the stored coefficients, standard errors, confidence intervals and p-values (Appendix B gives the full cell-by-cell comparison). The other 267 of the 291 historical models (Section 5.7) passed internal design checks; apart from two unweighted counterparts verified in Appendix B, they were not re-estimated with a second library. This establishes that the reported numbers are what the documented specification, correctly implemented, actually produces — it is **not** a multiplicity correction. No formal adjustment for testing 24 cells (2 products × 3 outcomes × 4 sample groups) has been applied, so the check does not make this result, or its five other nominally significant siblings, any less likely to reflect chance given the number of cells examined, and it does not establish causality.

Weighted vs. unweighted, same observations. The pooled water–HAZ coefficient is sensitive to the equal-country weighting convention. It is 1.03 when weighted (equal country totals) and 0.41 [0.03, 0.80], $p = .035$, when unweighted on the identical 23,018 observations. This is a disclosed, authorised sensitivity to a stated design choice (Section 4.5), not evidence

that either weighting is more "correct", and both are reported together whenever this coefficient is cited. Equal country weight totals do not give each country equal influence on the pooled slope, which also depends on each country's residual exposure variation (Section 4.5). Neither the weighted nor the unweighted estimate is a population-representative estimate (Section 3.8).

Residual exposure variation and the 10-point scale. The pooled water–HAZ model's residual within-cluster exposure standard deviation after all controls is 0.43 percentage points. The conventional 10-percentage-point coefficient scale is therefore roughly 23 times larger than the exposure contrast that actually identifies the model. One residual standard deviation of exposure corresponds to approximately 0.044 HAZ standard deviations. The 10-point coefficient should therefore be read as a reporting convention for comparability across studies, not as a description of a change that is typically observed within a cluster.

5.7 Alternative windows and common-sample comparison

The 291-model register (`docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`) decomposes exactly into: 24 primary established-date models; 24 unweighted-comparison models; 72 alternative-window models (the window choice examined here: three further windows × 2 products × 3 outcomes × 4 sample groups); 24 wealth-adjustment, 24 containing-pixel, 24 rural-10km, 48 stricter-completion-rule (two postnatal windows), and 24 calendar-age-construction models; 18 Ethiopia-provisional models; and three 3-model overlap-exclusion families (9 total) — summing to 291. A separate 192-row file re-presents the window comparison on two sample bases (each window's own eligible sample and the sample common to all four); its own-eligible-sample rows overlap with, rather than add to, the 291 already counted. To avoid repeating every cell, only the pooled water–HAZ cells are shown below as the headline case; the corresponding sanitation and WAZ/WHZ sensitivity cells are neither more nor less favourable to a WASH association (most show the same wide, zero-spanning intervals as Section 5.6) and are reported in full in the public aggregate tables, omitted here for brevity, not because water–HAZ happened to be significant.

Table 5.8 — Historical extension: alternative exposure-window and common-sample comparison (pooled water–HAZ).

Window	Pooled water–HAZ β [95% CI]	Sample
Post-12-month (main)	1.03 [0.51, 1.54]	Own-eligible, N = 23,018
Birth-year	0.54 [0.12, 0.96]	Own-eligible
Prenatal (9 months)	0.64 [0.24, 1.03]	Own-eligible

Window	Pooled water–HAZ β [95% CI]	Sample
Post-24-month	1.75 [0.81, 2.70]	Own-eligible
Post-12-month, common-sample	1.64 [0.68, 2.60]	All-window-eligible common sample, N = 16,419

Note. Coefficients are per 10 percentage points, as in Table 5.5. "Own-eligible" denotes each window's own eligible sample; the common sample contains only the children eligible under all four windows.

The first four rows each use their own window's eligible sample, so differences between them reflect differences in samples as well as in windows; only the last row holds the sample fixed. The post-12-month estimate changes from 1.03 on its own eligible sample to 1.64 on the smaller common sample eligible under all four windows. This is a **sample-composition effect** (children with complete 24-month windows differ systematically from the full post-12-month sample), not evidence favouring the validity of one window over another. No window was selected because of the significance of its coefficient; all four are reported together, as planned before any coefficient was examined (Appendix C).

5.8 Further sensitivity: wealth, spatial method, completion rule

Table 5.9 — Historical extension: further sensitivity of pooled water–HAZ to wealth adjustment, spatial method, completion rule and the exclusion of Ghana.

Sensitivity	Pooled water–HAZ β [95% CI]
Main (post_12m, weighted)	1.03 [0.51, 1.54]
Wealth quintile added	1.01 [0.50, 1.52]
Containing-pixel assignment (vs. buffer average)	0.68 [0.26, 1.10]
Rural 10 km buffer (urban unchanged)	1.01 [0.42, 1.60]
Stricter completion rule (+1 month)	0.95 [0.43, 1.47]
Excluding Ghana (DHS/IHME input overlap)	0.74 [0.24, 1.24]

Note. Coefficients are per 10 percentage points, as in Table 5.5. The row excluding Ghana pools Kenya and Nigeria only, and is not an overlap-free estimate: Kenya's own DHS survey also feeds into the IHME water product used as its exposure (Section 6.5), so excluding Ghana removes only one of the two established-date countries with this input overlap; every other row uses the three-country established-date sample.

Adding wealth leaves the coefficient essentially unchanged (1.01 vs. 1.03). This shows only that the reported conditional coefficient is not sensitive to this particular adjustment. It does not establish that wealth mediates the historical WASH exposure, and a small change under adjustment is not itself evidence against the competing possibility that wealth is a confounder the main specification should have included. Section 3.8 states both the mediator

concern and this competing concern, and this sensitivity adjudicates neither. Nor does it, on its own, support or contradict a cluster-fixed-effects absorption mechanism, which this thesis's own baseline model separately demonstrates is not how cluster fixed effects behave (Section 4.2). The spatial-method and completion-rule sensitivities move the coefficient by amounts smaller than, or comparable to, the gap between the main weighted and unweighted estimates, and none of them is privileged over the main specification.

5.9 Ethiopia: provisional, excluded from the established-date pooled sample

Ethiopia's three interview-date candidates are not treated as equally valid sensitivity evidence. Candidate C1 uses a 30-day Ethiopian-calendar arithmetic conversion that has already been shown, independently of the WASH analysis, to be invalid for the delivered recoded date fields (Appendix C). It is excluded from this section and from every reporting table, and its past numerical agreement with the other two candidates is not treated as evidence of robustness. Candidates C2 and C3 remain provisional, since neither has been independently validated. They agree closely with each other (water–HAZ: −0.39 for both; sanitation–HAZ: −0.45 for both; all four confidence intervals include zero). These estimates are reported here and in Appendix C only as an explicitly exploratory sensitivity analysis. They are never pooled into, or compared with, the established-date headline results above.

6 Discussion and limitations

6.1 Relating the findings to the literature

The baseline findings are consistent with the existing literature, but only narrowly. Headey and Palloni (2019) find that, under a within-region panel design netting out time-invariant confounding, sanitation improvements predict large reductions in diarrhoea and mortality but are **not** associated with stunting or wasting, while improved water is statistically insignificant for most outcomes except piped water (reduced stunting). Their null sanitation–stunting finding is a close, specific parallel to this thesis's own null household- and community-sanitation associations with HAZ (Sections 5.1–5.2): both find no detectable sanitation–growth association once a design better suited to netting out confounding is used, each in its own way (region-level panel fixed effects there; region or cluster fixed effects here, with no household-level fixed effect — household WASH is the exposure, not the FE grouping). The route to the null differs by

exposure level here: community sanitation is strongly associated with HAZ before adjustment but loses that association once socioeconomic controls and region FE are added (Table 5.2), while household sanitation is never significant (Table 5.1). This is therefore a narrower, more defensible claim than saying WASH–growth associations "generally" attenuate under better controls, which would overstate what any single study, this thesis included, establishes on its own.

The household cluster-fixed-effects associations here are not significant at the 5% level (water–HAZ $p = .062$; sanitation–HAZ $p = .548$), broadly consistent with the strongest randomised evidence available: no WASH-only arm in the three trials showed a statistically detectable growth effect (Section 2.3), while the nutrition arm (no WASH component) showed a modest positive effect, and adding WASH to nutrition brought no additional benefit over nutrition alone (Cumming et al. 2019). This should not be overstated as confirmation: this thesis's exposure is a facility-type classification in cross-sectional data, not a randomised intervention with prospective follow-up from near birth, so the two literatures answer related but not identical questions, and similarity of findings does not validate this thesis's coefficients. The one significant household-level result (household water conditional on community water and the other two terms, HAZ $p = .043$; Section 5.3) is negative and just below threshold — one cell among twelve in that table, not treated as evidence that improved household water harms growth; no mechanism in Section 2 supports that reading, and this thesis proposes none.

The historical extension's pooled water–HAZ association is positive and significant: 1.03 per 10 percentage points, $p < .001$, or 0.41 [0.03, 0.80], $p = .035$, when the same observations are unweighted (Section 5.6). The coefficient was independently reproduced with a second estimation library on a separately rebuilt sample, as were all 24 primary established-date cells. The other 267 of the 291 historical models passed internal design checks and, apart from two unweighted counterparts, were not re-estimated with a second library (Section 5.6, Appendix B). Computational reproduction is not a multiplicity correction. No formal adjustment has been applied for the number of cells estimated across the baseline and historical analyses, so this thesis has neither quantified nor ruled out the possibility that this result is one significant draw among many tested cells. It would be tempting to read the coefficient as evidence that introducing a time dimension recovers an association that the single-round baseline design cannot detect. Section 6.2 explains why this thesis does not draw that conclusion.

6.2 Why the baseline and historical results cannot be compared as a test of exposure timing

This point has its own subsection because it is the most tempting overinterpretation that the structure of this thesis invites. The research question of the historical extension (Sections 1.2 and 1.3) is framed deliberately to avoid it. The extension asks whether modelled early-life local WASH coverage is associated with outcomes *within its own sample*, not whether assigning a historical rather than a current exposure changes an otherwise identical estimate.

The baseline and historical samples differ in at least four respects at the same time, not in one:

1. **Different children and survey rounds.** The baseline uses 2022–2025 DHS rounds from four countries. The historical extension uses 2014–2018 rounds from three; Ethiopia's 2016 round enters only as a provisional sensitivity analysis outside the established-date pooled sample (Section 1.4). No child appears in both samples.
2. **Different exposure definitions and construction.** The baseline household exposure is the survey response of each child's own household at the survey date. The community exposure is an unweighted leave-one-out proportion among the other sampled, classifiable HR household survey responses in the same cluster, not a census of every household in the cluster (Section 3.3). The historical exposure is a modelled, area-averaged, annual gridded coverage estimate from an external source (IHME), matched to a specific postnatal window (Sections 3.5 to 3.7). These are not the same quantity measured at two different times; they are two different kinds of measurement.
3. **Different eligibility and completion rules.** The historical sample also requires complete spatial coverage and a complete exposure window verified against the interview date (Section 3.7). These restrictions have no counterpart in the baseline sample.
4. **Different control sets and identifying variation.** The main baseline household specification uses cluster fixed effects with household- and individual-level socioeconomic controls, including maternal age and education, literacy, household size and wealth quintile (Section 3.4). In the baseline, region fixed effects appear in the household region-fixed-effects comparison (Section 4.1), in the separate community specification (Section 4.3) and in the joint household–community model (Section 4.4); they are never stacked with the household model's cluster fixed effects. The historical specification uses cluster fixed effects with country-by-birth-year terms and excludes wealth from the main specification, for a reason specific to the historical design (Sections 3.8 and 4.6). The historical cluster-fixed-effects model is identified by genuine within-cluster variation

across birth cohorts (Section 4.6). The baseline community model, by contrast, is deliberately *not* estimated with cluster fixed effects, for a reason specific to its leave-one-out construction (Section 4.3) that has no bearing on the historical design.

Because all four differ at once, a contrast between the baseline's largely null associations and the historical extension's selected pooled water–HAZ headline cannot be cleanly attributed to "the effect of assigning historical instead of current exposure" — that headline is one of six nominally significant cells among the 24 primary cells, not the only significant historical result (Section 5.6). The contrast could instead, or in addition, reflect different confounding structures across the two country sets, different measurement-error properties of a modelled surface versus a survey response, the different control sets, or chance given the number of cells estimated (the multiplicity point of Section 5.6 applies here too). No formal test of the baseline-versus-historical difference was carried out, and a significant coefficient beside non-significant ones does not by itself show the two differ significantly. This thesis does not adjudicate between these explanations: its design, two separately constructed samples rather than one shared sample in which only exposure timing varies, cannot do so.

What the within-historical-sample window comparison does and does not hold constant. Section 5.7 compares four exposure windows (birth year, prenatal, post-12-month and post-24-month) within the *same* historical country set and estimation approach. The comparison holds constant the survey rounds, the exposure source and spatial-extraction method, the control set and the estimator. In terms of specification, it changes only which months of the IHME surface are averaged into each child's exposure value. It does **not** hold the estimation sample constant across the four windows: each window has its own eligible sample, which differs according to which children have complete coverage for that specific window (Section 5.7). Table 5.8 shows how much this can matter. With the window held at post-12-month, the estimate moves from 1.03 on its own eligible sample to 1.64 on the smaller sample eligible under all four windows, a change in sample composition rather than in the window. Differences between the own-sample window estimates therefore combine window changes and sample-composition changes. The window comparison also says nothing about the baseline-versus-historical contrast discussed above, and this paragraph keeps the two separate.

6.3 Limitations: unresolved documentation

Two issues in the historical extension remain genuinely unresolved documentation gaps; they are neither design choices nor inherent limitations. First, no DHS or IPUMS source could be located that documents a measurement-subsample or anthropometry-specific sampling weight,

despite repeated attempts with the DHS Biomarker Manual and alternative official sources (Appendix A records the exact sources tried). This thesis's working inference is that within-survey PERWEIGHT normalisation removes a constant measurement-subsample selection factor, under an assumption of random household-level selection. The inference is stated as such, not as a documented fact, and it cannot correct for differential non-response among children who were eligible but not measured. Second, the gap between Nigeria 2018's official count of measured children (12,806) and the number of children with non-missing anthropometry in the IPUMS birth-history extract (11,704) remains unexplained (Section 3.9).

Both points bear on which children the historical estimates describe, so this thesis reports its historical coefficients as **weighted conditional associations among eligible measured children**, not population-representative estimates. Reporting the unweighted estimate alongside (Section 5.6) does not resolve this, since the selection question is about who was measured at all, not how measured children are weighted relative to one another; if measurement or non-response related to both local WASH coverage and growth, the estimated associations could differ from the full eligible population, and the data here cannot show whether, or in which direction, they do.

6.4 Limitations: authorised design choices, disclosed and tested

Three features of the design are deliberate, disclosed choices, not errors.

First, the equal-country-total pooled weighting (Section 4.5) gives each established-date country equal total observation weight in a single pooled common-slope regression, not a population-size-weighted estimate, and not the same as averaging each country's own separately estimated coefficient: statistical influence on the pooled slope also depends on residualized exposure variation after controls, so equal weight totals do not imply equal influence. This choice's consequence is disclosed alongside the main estimate throughout: the pooled water–HAZ coefficient moves from 1.03 (weighted) to 0.41 (unweighted, same observations); neither is population-representative (Section 3.8).

Second, the working 12-month postnatal exposure window was chosen during the design phase, before any model was estimated, for a documented reason: it is an early-life window whose completion can be checked against the interview date. It is nowhere described in this thesis as "preregistered". The literature basis for this specific window length is modest. Cumming and Cairncross (2016) support early-life exposure in general and, if anything, lean toward a 24-month window, which is why all four candidate windows (Section 5.7) are reported together.

Third, wealth quintile is excluded from the historical extension's main specification because it may be a post-exposure mediator of historical WASH conditions and because it is correlated with the WASH exposure itself (Section 3.8). This is a substantive identification concern. It is not a claim that cluster fixed effects remove wealth mechanically, a claim that the baseline model shows to be false by including and estimating wealth directly (Section 4.2). Section 3.8 also states the competing concern: current wealth may proxy earlier socioeconomic circumstances that confound the association, in which case excluding it leaves that confounding unaddressed. Adding wealth leaves the pooled water–HAZ coefficient essentially unchanged (1.01 vs. 1.03; Section 5.8), but this sensitivity does not adjudicate between the two readings.

6.5 Limitations: inherent, requiring disclosure rather than resolution

DHS/IHME input overlap affects two of the three established-date historical countries, Ghana and Kenya, whose own DHS surveys feed into the IHME gridded estimates used as the exposure. The exposure measure is therefore not fully independent of the survey data used in the analysis. This thesis reports the overlap-exclusion sensitivity alongside the main estimate rather than treating the overlap as removable: the pooled water–HAZ estimate excluding Ghana is 0.74 (Section 5.8). Because that estimate pools Kenya and Nigeria, Kenya's overlap remains in it, so the sensitivity addresses only part of the overlap.

Current residence, as recorded at the survey date, is used as a proxy for residence during each child's early-life window. Children whose survey residence differs from their early-life residence may be assigned an exposure that does not represent their early-life location, under both the primary buffer-average method and the containing-pixel sensitivity. This is a form of exposure measurement error whose extent in these samples is unknown.

DHS cluster coordinates are randomly displaced for confidentiality (Section 3.4), and this limitation holds whatever extraction method is used. The area-weighted buffer average, this thesis's primary method, reads pixels around the displaced coordinates, not the true ones. It is an assignment choice: it does not recover the true location, it does not validate or correct for the displacement that DHS actually applied to each cluster, which is not disclosed, and it is not a population-weighted estimate (Sections 3.5 and 3.6). The containing-pixel sensitivity (Section 5.8) reads a single displaced pixel instead, which is subject to the same location uncertainty through a different mechanism. Neither method is immune to displacement, and neither resolves it. Under its primary method, this thesis's spatial extraction computes an area-weighted average of the raster surface within each buffer, which is a different quantity from

the source study's population-weighted administrative aggregate (Section 3.5).

The IHME exposure surface is itself a *modelled* estimate with its own uncertainty, not a direct observation, and its supplied lower and upper bounds are not used as regression inputs, so the standard errors of the historical coefficients do not reflect uncertainty in the exposure surface itself. The surface is also available only at annual resolution: for the final calendar year a window touches, the annual value used can include months after the child's own interview date even once that window's postnatal completion rule (Section 3.7) is satisfied. The project's construction flags every such case but neither excludes nor adjusts for it, so this thesis does not claim every historical exposure value is measured strictly before the outcome it is matched to.

The residual within-cluster exposure variation that actually identifies the historical coefficients is small: 0.20 to 0.88 percentage points, depending on country and product (Section 4.6), and 0.43 percentage points for the pooled water–HAZ model (Section 5.6). This is modest relative to the 10-point scale used for reporting, so a coefficient per 10 percentage points is a reporting convention, not a description of a change typically observed within a cluster (Section 5.6). Finally, Ethiopia's interview-date uncertainty (Section 5.9; Appendix C) is treated as an inherent limitation of that round's recoded date fields, not as a design choice made in this thesis. Its estimates remain provisional and outside the established-date pooled family.

6.6 What this thesis does not claim

This thesis does not claim that its findings establish a causal effect of WASH facilities on child growth, in either direction. Both analyses are cross-sectional and observational, and fixed effects remove only what is shared within a region or cluster; confounding that varies within clusters may remain (Section 4.2). The thesis does not claim that its null or weak baseline associations are evidence that WASH investment is an ineffective use of development resources: non-significance here means that the evidence in this sample and design is inconclusive, not that no association exists. It does not generalise findings from Ethiopia, Ghana, Kenya and Nigeria, four countries selected for data availability and recency (Section 1.4), to Sub-Saharan Africa as a region. It does not treat the historical extension's computational verification (Appendix B) as evidence of improved causal identification. And, as Section 6.2 sets out, it does not treat the difference between the baseline's largely null results and the historical extension's selected pooled headline result as evidence that exposure timing explains that difference, because the two samples differ in several other respects that the design cannot separate from timing. With these boundaries stated, Section 7 draws together what the evidence of this thesis does, modestly, support.

7 Conclusion

This thesis examined the association between water and sanitation conditions and the anthropometric outcomes of children in four Sub-Saharan African countries: Ethiopia, Ghana, Kenya and Nigeria. Two distinctions ran through the analysis. The first is between a child's own household facilities and the WASH conditions of the surrounding community. The second is between a single exposure measured at the survey date and a preliminary, geospatially matched measure of modelled local WASH coverage during the child's own early-life window. The second measure was constructed for an earlier survey round in three of the four countries, with Ethiopia retained only as a provisional sensitivity analysis (Section 1.4).

The baseline evidence does not show a clear, robust association between improved household WASH and child height-for-age, weight-for-age or weight-for-height when households are compared within the same DHS cluster. None of the six household cluster-fixed-effects coefficients (three outcomes, two exposures) reaches significance at the 5% level under this thesis's primary inference convention; household water–HAZ, with $p = .062$, crosses only a 10% threshold (Section 5.1). Under this design, that is an inconclusive result, not evidence that no association exists.

The community measure answers a related but distinct question. It is a leave-one-out coverage rate among the other sampled households in the cluster, and it is estimated with region rather than cluster fixed effects (Sections 4.3 and 5.2). Community coverage shows a large, strongly significant association before adjustment for most outcome and exposure pairs. Weight-for-height water is the exception: it is small and non-significant from the start. Once socioeconomic status and region fixed effects are added, the association is no longer statistically significant for any pair. This attenuation is consistent with confounding by local development, but the design cannot establish confounding as the only explanation.

These baseline findings are broadly consistent with the strongest available causal evidence on WASH interventions. In the three major trials, the basic, household-level WASH interventions tested showed no statistically detectable effect on linear growth, and the trial authors state explicitly that this does not challenge the biological plausibility of WASH affecting growth (Cumming et al. 2019). The parallel is suggestive rather than confirmatory, because the exposure in this thesis is a facility-type classification in survey data, not a randomised, comprehensive intervention package (Section 2.3).

The historical extension links earlier DHS rounds from Ghana, Kenya and Nigeria to IHME's modelled annual local WASH coverage for a defined early-life window. Its exposure is an area average of modelled coverage around displaced cluster coordinates; it is neither

observed historical household access nor population-weighted coverage (Sections 3.5 and 6.5). The selected headline cell is the pooled established-date association between local water coverage and height-for-age: $\beta = 1.03$ per 10 percentage points (95% CI [0.51, 1.54], $p < .001$). It is not the only nominally significant historical result. Six of the 24 primary cells have $p < .05$, one of them a negative Ghana sanitation cell (Section 5.6). The headline cell was reconstructed independently with a second estimation library on a separately rebuilt sample, as were all 24 primary cells; the other 267 of the 291 historical models passed internal checks but, apart from two unweighted counterparts, were not re-estimated in this way (Appendix B). This reproduction confirms the computation. It does not adjust for the number of cells tested, and it does not make the association causal.

The headline finding carries the qualifications set out in Section 6. It depends on the pooled weighting convention: on the same 23,018 observations, the unweighted estimate is $\beta = 0.41$ (95% CI [0.03, 0.80], $p = .035$), and equal country weight totals do not give the three countries equal influence on the pooled slope (Section 5.6). The weighting of the measured subsample relies on a working inference, because no official documentation of an anthropometry-specific sampling weight could be located (Section 6.3). The estimates are therefore weighted conditional associations among eligible measured children, not population-representative effects. The exposure source draws on the DHS surveys of Ghana and Kenya themselves, current residence stands in for early-life residence, and the residual within-cluster exposure variation that identifies the model is much smaller than the 10-point reporting scale (Section 6.5). Ethiopia's estimates remain provisional and outside the established-date pooled family. Finally, the contrast with the baseline results cannot be attributed to exposure timing, because the two samples differ at the same time in their children and survey rounds, exposure construction, eligibility rules and control sets (Section 6.2). The historical result is a preliminary, qualified finding rather than a settled one, and this thesis does not use it to overturn or reinterpret the more cautious baseline conclusions.

Taken together, the thesis makes two separate contributions in two separate samples (Section 1.5). In the baseline sample, it compares own-household and leave-one-out community WASH exposure side by side, within one four-country sample and one consistent set of controls, with each exposure estimated under the fixed-effects structure its construction requires. In the historical sample, it makes a documented, preliminary attempt to introduce genuine temporal variation in early-life exposure through geospatially matched coverage data. Because the historical data cannot separate a child's own household from its community, this second contribution does not extend the household-versus-community comparison.

Neither set of results, alone or together, establishes a causal effect of WASH facilities on

child nutritional outcomes in these four countries. Fixed effects remove only what is shared within a region or cluster, and the four countries were selected for data availability and recency, not to represent Sub-Saharan Africa as a whole (Sections 1.4 and 6.6). Nor can the results be read as evidence for or against the value of WASH investment as a development policy. Weak or inconclusive nutritional associations in an associational, four-country design do not show that such investment lacks value, and no causal policy effect or cost-effectiveness was estimated. What the thesis offers instead is an account of where its evidence is stronger, where it is weaker, and where further data or a different design would be needed before a stronger claim could responsibly be made. The two clearest gaps are the unresolved measurement documentation of the historical extension and the exposure-timing limitation inherent in the baseline design.

Future work could address these gaps in three ways that follow directly from the limitations above. First, validating the Ethiopian interview-date conversion would allow the historical approach to cover all four countries on an established-date basis (Appendix C). Second, the measurement-subsample weighting question could be resolved through direct engagement with DHS methodological documentation. Third, geospatial WASH matching could be combined with a design that follows the same geographic units over successive survey rounds (Section 1.2). These would be natural next steps toward narrowing the gap that this thesis has described but not closed.

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Appendix A — Variable definitions, sample flow, and authoritative sources

A.1 Variable definitions

Baseline and historical variables are listed in separate tables, because the two samples are built by different source pipelines (country-specific DHS recode files for the baseline; IPUMS-DHS for the historical extension) and use different native field names for the same kind of quantity. A baseline field is never the same field as its historical counterpart under another name, even where the two serve an analogous purpose (Section 3.4, Section 3.8).

The unit of analysis in both samples is the child. Scripts named in Table A.1 are in `scripts/dhs_harmonization/` and those in Table A.2 in `scripts/historical_wash/`; all paths are relative to the repository root.

Table A.1 — Baseline variables (Sections 3.1–3.4)

Variable	Definition	Source file (authoritative)
HAZ, WAZ, WHZ	Height-for-age, weight-for-age and weight-for-height z-scores, WHO Child Growth Standards reference (WHO Multicentre Growth Reference Study Group 2006). Read from the KR fields `hw70`, `hw71` and `hw72`, which store each z-score multiplied by 100; every valid value is divided by 100 once, giving the analysis scale in standard deviations. A value is valid unless it is system-missing or one of the DHS special codes 9996 (height out of plausible limits), 9997 (age in days out of plausible limits), 9998 (flagged cases) or 9999 (missing). All five non-valid states become missing, never a number. The bounds −6 to 6 (HAZ) and −6 to 5 (WAZ, WHZ) are a validation check that stops the pipeline if a valid value falls outside them; they do not reclassify any value.	DHS Children's Recode (KR); `02_load_flag_anthro.py`
Household water (binary)	1 if the household drinking-water source on the child's KR record (`v113`) is JMP-improved; 0 if it is unimproved or surface water. Code 96 (other; "Unknown") and code 97 (not a de jure resident; "structurally not applicable") are set to missing, not to 0. "Improved" is a facility-type category; it is not a measure of a safe or safely managed service.	`01_wash_mapping.py`, `03_build_indicators.py`, `12_robustness.py`

Table A.1 (continued)

Variable	Definition	Source file (authoritative)
Household sanitation (binary)	1 if the KR sanitation facility (`v116`) is JMP-improved, including an improved facility shared with other households (the Basic and Limited service levels derived from `v160` are both counted as improved); 0 if it is unimproved or open defecation. Code 96 ("Unknown"), code 97 ("structurally not applicable"), an improved facility whose sharing code is 7 (also structurally not applicable) and Ghana's flush-to-biodigester code 16 (unclassified in the core specification) are set to missing.	`01_wash_mapping.py`, `03_build_indicators.py`, `12_robustness.py`
Community water/sanitation (`water_rate_loo`, `sanitation_rate_loo_core`)	Unweighted leave-one-out proportion of improved facilities among the *other* sampled, classifiable Household Recode (HR) households in the child's DHS cluster, classified from `hv201`, `hv205` and `hv225` with the same mapping as the household variables. The child's own household is excluded from both the numerator and the denominator, so all children in one household receive the same value. Households recorded as Unknown or structurally not applicable, and, in the core sanitation variant, households using Ghana's unclassified biodigester, are also excluded from both. The value lies on the 0–1 scale and is missing when no other classifiable household remains in the cluster. HR covers sampled households only, so the measure is not a census of every household in the physical community, and it is not an average over children. A robustness variant, `sanitation_rate_loo_robustness_biodigester`, counts code 16 as improved; it differs from the core variant only in Ghana.	HR; `08_hr_cluster_wash_exposure.py`, `01_wash_mapping.py`
`v005` → `weight_original`	The woman's individual sample weight from the DHS Children's Recode (KR), divided by the documented factor of 1,000,000. Each child's KR record carries the mother's weight, because DHS computes no separate per-child weight within KR. The distinct Births Recode (BR) is not used. This is **not** the household weight `hv005`, which exists only in the Household Recode (HR) and is not used in the baseline models. Pooled models rescale `weight_original` to equal country totals within each model's estimation sample (Section 4.5); country-specific models use it unchanged.	`04_build_country_kr.py`; pooled rescaling in `11_main_regressions.py`
`v021` → `psu` → `country_psu`	DHS's native primary sampling unit code, copied to `psu` and combined with a country prefix into the country-qualified cluster key `country_psu`. Pooled models use `country_psu` for cluster fixed effects and clustered standard errors; country-specific models cluster on `psu`.	`04_build_country_kr.py`, `05_pool_countries.py`

Table A.1 (continued)

Variable	Definition	Source file (authoritative)
`b19` (child's own age in months)	The DHS recode's standard current-age-in-months field, stored unchanged as `b19_raw` and used uniformly for all four baseline countries (no country-varying construction); a linear control from Model 2 onwards.	`04_build_country_kr.py`, `11_main_regressions.py`

Note. Baseline source files: DHS KR and HR recodes for Ethiopia 2024–25, Ghana 2022, Kenya 2022 and Nigeria 2024; the baseline does not use IPUMS-DHS. The household and community WASH constructions are also described in `docs/empirical_strategy.md` §2. The pipeline applies the same special-code list to all four countries; whether DHS's own criteria behind code 9998 are identical across these surveys is not established here. Community exposure groups HR households by the HR cluster number `hv001` and links them to children through `v001` and `v002`, whereas cluster fixed effects and clustered standard errors use `v021`. Whether these identifiers correspond one-to-one in every baseline country has not been checked against the restricted data and remains an open verification question.

Table A.2 — Historical-extension variables (Sections 3.5–3.8)

Variable	Definition	Source file (authoritative)
`IDHSPSU`	Sample-qualified DHS cluster code from IPUMS-DHS, unique across survey rounds; used for cluster fixed effects and clustered standard errors. **It is the historical extension's own cluster identifier, distinct from the baseline's `country_psu` above.**	IPUMS-DHS / DHS recode; `16_estimate_preliminary.py`
`PERWEIGHT`	The IPUMS-DHS standard sampling weight for child-level tabulations, rescaled to equal country totals within each model's final estimation sample (Section 4.5). **It is the historical extension's own weight field, distinct from the baseline's `weight_original` above.** It is not a documented anthropometry-specific weight (Section A.4). `KIDWT`, an IPUMS population adjustment factor, is not used.	IPUMS-DHS documentation; `16_estimate_preliminary.py`
W_IMP, S_IMP	IHME modelled annual local coverage of JMP-improved water and sanitation: percent of the population with access, on a 0–100 scale, ~5×5 km grid, 2000–2017. Each child's value is the area average of the MEAN surface within a buffer of 2 km (urban) or 5 km (rural) around the displaced cluster coordinates; it is not population-weighted. Containing-pixel assignment and a rural 10 km buffer are sensitivity methods.	`docs/provenance/historical_wash_ihme_product_definitions_v1_2026-10-07.md`; `10_spatial_extraction_v3.py`

Table A.2 (continued)

Variable	Definition	Source file (authoritative)
`age_m` (child's own age in months)	**The historical extension's own age field, distinct from the baseline's `b19` above.** Reported completed months (`KIDCURAGEMO`) where the source delivers them (Ethiopia, Nigeria); otherwise the documented calendar-month difference between interview and birth dates (Ghana, Kenya).	`16_estimate_preliminary.py`, `14_build_analysis_samples.py`
`b_year` (birth year)	The child's birth year, from the birth-date century-month code; used for survey-qualified country-by-birth-year terms in the historical models. Derived in the analysis-sample construction stage and confirmed independent of the Ethiopian interview-date candidate (Appendix C).	`14_build_analysis_samples.py`
`cluster_key`	A copy of `IDHSPSU` (`cluster_key = IDHSPSU`), which already encodes survey and cluster, so the key is sample-qualified by construction.	`14_build_analysis_samples.py`; stored in `data/processed/analysis_samples/run_20261006T184744Z/analysis_candidates_RESTRICTED.parquet` (RESTRICTED; schema only cited here)

Note. Historical source: the IPUMS-DHS child-level extract for Ghana 2014, Kenya 2014 and Nigeria 2018 (the established-date family); Ethiopia 2016 enters only as a provisional sensitivity analysis (Appendix C). Historical HAZ, WAZ and WHZ come from the IPUMS fields `HWHAZWHO`, `HWWAZWHO` and `HWWHZWHO` (stored multiplied by 100 and divided by 100 once); code 9999 marks a child outside the measurement universe, and codes 9995–9998 and system-missing values are not valid. IHME raster values are percentages; NoData pixels are excluded from the valid area rather than treated as zero. A child-year counts as complete only when the valid area covers the buffer to within 10^{-3}; partial-coverage values are kept for diagnostics only. The supplied LOWER and UPPER surfaces are not regression confidence intervals and do not enter the models. Exposure windows (`12_construct_exposure_windows.py`): the birth-year window is the full calendar year of birth; the prenatal window is the nine months before the birth month (birth month excluded); the post-12- and post-24-month windows start with the birth month (included). Each calendar year is weighted by its share of the window's months, and a window needing any year outside 2000–2017 is not assigned a value. Only the two postnatal windows require every window month to have ended before the interview month; the prenatal and birth-year windows have no interview-date completion check. Because the surface is annual, a value may include months after the interview; such cases are flagged, not excluded (Section 3.7).

A.2 Baseline sample flow

Counts are unweighted numbers of children.

1. Pooled harmonised child file: 70,231 children, 4,481 clusters and 114 regions.
2. Outcome-valid samples within the measurement universe: 36,985 children for HAZ, 37,260 for WAZ and 37,088 for WHZ, in 4,434 clusters. Validity is the outcome rule in Table A.1; the code applies no separate measurement-universe filter. No child is lost at the exposure or covariate step.
3. Common samples requiring both household and community exposures to be classifiable: 36,040 (HAZ), 36,308 (WAZ) and 36,145 (WHZ) children, in 4,426 clusters and 25,627, 25,759 and 25,665 households.

The baseline code contains no singleton-cluster exclusion step (Section 4.5), so no baseline singleton count exists.

Full flow: `outputs/household_community_wash/01_development_sample_flow.csv` and, for the outcome-valid samples, `outputs/tables/step11_model_sample_flow.csv`.

A.3 Historical sample flow (established-date, post-12-month window, water, HAZ, as a worked example)

The example is the primary pooled model (Ghana 2014, Kenya 2014, Nigeria 2018; W_IMP; primary buffer method; weighted).

1. Raw analysis candidates in the three established-date samples: 60,772 child records (Ghana 5,884; Kenya 20,964; Nigeria 33,924).
2. Candidate label `GREGORIAN_ESTABLISHED`: 60,772 (no further loss for the three established countries).
3. Complete postnatal window and complete spatial coverage: 42,052.
4. HAZ valid and within the measured universe: 23,135.
5. Complete cases on covariates, exposure, weight and cluster key: 23,135 (no further loss).
6. Singleton-cluster exclusion: −117 clusters and −117 observations.
7. **Final: 23,018 children in 3,114 clusters** (Ghana 1,940 in 353; Kenya 14,166 in 1,498; Nigeria 6,912 in 1,263).

Singleton clusters are removed separately from each model's own final sample, so the number removed differs between models; the three country models remove 27, 16 and 74. Section 3.9 explains why the Nigeria counts differ from the official 2018 anthropometry count; that gap remains unresolved.

Steps 1, 2, 4, 6 and 7 can be checked against two public aggregate files in `docs/provenance/`: `historical_wash_analysis_samples_v1/sample_flow_AGGREGATE.csv` and `historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`. Step 3 is recorded only in the restricted log, because the public flow file assigns each excluded record to its first failing condition in a different order. The steps were reproduced independently in the execution log of `scripts/historical_wash/20_validate_pooled_estimate.py` (`data/processed/estimation/pooled_independent_validation_20261007T090003Z/execution_log.txt`, RESTRICTED).

A.4 DHS/IPUMS official documentation sources attempted for the measurement-subsample weight (Section 3.8, Section 6.3)

Table A.3 — Official documentation sources consulted for a measurement-subsample weight

Source tried	Result
DHS-8 Biomarker Manual (dhsprogram.com PDF)	HTTP 403, tried twice across two sessions
Guide to DHS Statistics (dhsprogram.com)	HTTP 403
World Bank microdata catalogue, child anthropometry variable page	Accessible; no weighting guidance found
Nigeria 2018 NDHS final report	Accessible; describes sampling weights by stage and cluster and a domestic-violence special weight, but no anthropometry- or biomarker-specific weight
IPUMS-DHS PERWEIGHT and KIDWT documentation pages	Accessible; confirms PERWEIGHT as the documented child-level tabulation weight; KIDWT is a population factor and is not used

No source documents a measurement-subsample-specific weight. This is recorded as an unresolved documentation gap (item 1 of `docs/provenance/historical_wash_issue_register_v1_2026-10-07.md`). No substitute weight or correction was invented; Section 3.8 states the working inference used instead.

A.5 Full authoritative-file index

The current index of baseline and historical output files cited in this thesis, with their status labels (validated for main reporting, validated sensitivity, exploratory, invalid or superseded, not estimable), is `docs/provenance/final_empirical_package_v3_2026-10-07/00_results_index.md`, together with `02_core_reporting_table.md` in the same folder. That index refers to `final_empirical_package_v2_2026-10-07/00_results_index.md` for the baseline entries and to `01_final_reporting_map.md` for the status of each result.

Appendix B — Model specifications, inference conventions, and independent verification

The estimating equations are stated in Chapter 4 (Sections 4.1–4.6). This appendix records how they are implemented for inference and how far the implementation has been verified. It covers the baseline household cluster-fixed-effects model (Section B.1), the historical extension (Sections B.2–B.4) and the estimator safeguard added after an external audit (Section B.4). Figures marked RESTRICTED come from execution logs and output folders that contain or derive from restricted microdata; they are carried here from those records and from the public provenance documents named in the text. They were not re-executed in preparing this appendix.

B.1 Inference-convention comparison (baseline household cluster-FE model, HAZ)

Table B.1 — Household water and sanitation coefficients for HAZ under the two inference conventions (baseline household cluster-fixed-effects model)

Convention	Water β	SE	p	Sanitation β	SE	p
PRIMARY (same-group CRV1; this thesis's main convention)	−0.085	0.046	.062	0.026	0.043	.548
FULL_DUMMY_DF sensitivity	−0.085	0.049	.080	0.026	0.046	.574

Note: Household water and sanitation are entered together; N = 36,040 children in G = 4,426 PSUs (`country_psu`). Both rows use the normal reference distribution for *p*-values and confidence intervals. Source: `outputs/household_community_wash/03_household_cluster_fe.csv`, rows `natural_sample_PRIMARY_SAME_CLUSTER_CRV1` and `natural_sample_FULL_DUMMY_DF_SENSITIVITY`, outcome `haz`.

The two conventions differ only in whether the small-sample CRV1 correction also penalises degrees of freedom for the absorbed cluster dummies. FULL_DUMMY_DF does; PRIMARY does not, following the project's cross-validated adjudication against `linearmodels.PanelOLS(auto_df=True)` (Sheppard 2024) (see `docs/provenance/cluster_fe_inference_adjudication.md`). Only standard errors, confidence intervals and *p*-values differ; the point estimates remain unchanged. The adjudicated PRIMARY convention is the one presented in the abstract, the introduction and the conclusion. The FULL_DUMMY_DF sensitivity is documented here and reported next

to the primary values in Section 4.5 and Section 5.1, where it is named explicitly. This does not mean that both conventions are repeated wherever else this coefficient is mentioned.

How the two conventions are computed. The model is fitted as weighted least squares on data demeaned within each PSU (weighted within-cluster demeaning, which uses only within-cluster variation and is equivalent to including a full set of PSU dummies), with `statsmodels` `cov_type="cluster"` clustered on the same `country_psu` grouping. The library's correction is

$$c = \frac{G}{G-1} \times \frac{N-1}{N-K},$$

where K is the number of columns in the design matrix it receives. Under PRIMARY that is $K = 18$: the household exposures and covariates after demeaning, with no constant and no PSU dummies. FULL_DUMMY_DF counts every absorbed PSU dummy as an estimated parameter, $K = 18 + 4{,}426 = 4{,}444$, which multiplies the PRIMARY standard error by $\sqrt{(N-18)/(N-4{,}444)}$ (1.0677 for HAZ) and recomputes p-values and confidence intervals from the normal distribution. The executed run log records `use_t: False` for these fits, so the PRIMARY p-values are normal-reference values as well (Section 4.5). The equivalence of demeaning and explicit PSU dummies was checked in the same script on a Ghana subset (N = 4,287, 613 PSUs): the two coefficients differ by 2.59×10^{-15} (water) and 5.57×10^{-15} (sanitation). The baseline code contains no singleton-cluster exclusion step, so G counts every PSU in the estimation sample (Section 4.5). The region-fixed-effects models are fitted differently: they use explicit region dummies, so K counts the constant, the covariates and the region dummies, and they too use the normal reference distribution. They are not part of Table B.1.

Why the baseline/`linearmodels` cross-validation agrees only to four or five decimal places, not further (2026-10-17 addendum). `docs/provenance/cluster_fe_inference_adjudication.md` records that an independent `linearmodels.PanelOLS(auto_df=True)` re-estimation of this model agreed with the native `statsmodels` standard errors "to four to five decimal places" but does not state whether that call also passed `group_debias=True`, the same opt-in flag whose omission is the entire explanation for the analogous, larger residual gap in the historical pipeline's own validation scripts (Section B.2 below). The call itself was not preserved as committed code, so this could not be settled by reading it. It was instead checked freshly in this pass: the PRIMARY household cluster-FE model (water and sanitation, HAZ) was re-estimated with `PanelOLS(entity_effects=True, drop_absorbed=True)` on the same sample and weights, first with `auto_df=True` alone, then adding `group_`

`debias=True`. Without the flag, the standard errors differ from the stored PRIMARY values by a relative 9.0e-05 (water) and 7.5e-05 (sanitation); with it added, the differences shrink to a relative 8.8e-06 and 2.5e-05. Both relative gaps are of the same order as the figure this appendix's own adjudication note anticipated, and adding the flag reduces each by roughly an order of magnitude, the same pattern Section B.2 documents for the historical pipeline. This supports, without proving beyond doubt for the specific unpreserved call, that the same omission explains the residual gap here. It changes nothing: the stored, published PRIMARY standard errors in Table B.1 come from `statsmodels`, not from this (or any) `linearmodels` cross-validation, and neither this check nor its predecessor altered any reported coefficient, standard error, confidence interval, or *p*-value.

B.2 Independent verification of the pooled historical model (Section 4.6, Section 5.6)

This thesis's own pipeline (`scripts/historical_wash/16_estimate_preliminary.py`) estimates all 291 historical models with a hand-written estimator based on weighted within-cluster demeaning. An earlier independent check (`scripts/historical_wash/17_validate_estimates.py`) re-estimated four representative *country-specific* cells: Ghana water–HAZ, weighted and unweighted; Ghana sanitation–WAZ; and Nigeria sanitation–HAZ. Three are primary cells; the unweighted Ghana cell (`m0005`) belongs to the unweighted comparison family. That script rebuilds each sample with its own filters, fits weighted least squares with explicit cluster dummies in `statsmodels`, and computes the CRV1 sandwich by hand with K counting only the non-fixed-effect regressors. Coefficients agreed within 3×10^{-14} (maximum absolute difference 2.96×10^{-14}), standard errors within a relative difference of 7.5×10^{-10}, and N and G were identical in all four cells (public output: `docs/provenance/historical_wash_preliminary_estimates_v1/independent_validation_PRELIMINARY_AGGREGATE.csv`).

That check did not cover the pooled multi-country model, which is the headline result of the historical extension. `scripts/historical_wash/20_validate_pooled_estimate.py`, written for this manuscript-preparation phase of the project, independently:

1. Rebuilds the estimation sample directly from the analysis-candidates stage (not from the stored, already-estimated model dataset), applying every filtering step explicitly and logging row counts (reproduced in Appendix A, Section A.3).
2. Estimates with `linearmodels.PanelOLS`, a different software library from the

project's own hand-written estimator, using entity (cluster) fixed effects and clustered CRV1 standard errors.

3. Reports both the PRIMARY (`auto_df=True`) and FULL_DUMMY_DF (`auto_df=False`) conventions.
4. Estimates the unweighted counterpart on exactly the same final observations.

The rebuild is independent in its filtering, weighting, design construction and estimation. It reads the same analysis-candidates file as the main pipeline, so the variables constructed upstream of that file are shared: the exposure assignment, the outcome validity flags, the birth year, the calendar age and the weight field. The check therefore does not re-verify those constructions.

First attempt and its correction, reported for transparency. An initial run of this independent script used a single pooled reference birth-year cell (rather than a separate reference year per country) and omitted the child's own age in months as a linear control. It produced β per 10 points = 1.385 against the stored 1.026, a discrepancy of roughly 35%. The sample size (N = 23,018), cluster count (G = 3,114) and exposure mean and SD nevertheless matched the stored values exactly, which pointed to a specification difference rather than a sample-construction error. After the documented design was matched exactly (country-local-reference birth-year dummies; child age in months included), the independent estimate converged to the values in Table B.2.

Table B.2 — Independent re-estimation of the pooled established-date water–HAZ model compared with the stored estimate

Quantity	Independent (corrected)	Stored (`m0004`)	Absolute difference
β per 10 points	1.0255531	1.025553067	6.7×10^{-8} against the six-decimal value 1.025553 (see note)
SE per 10 points (as originally run)	0.260694	0.260730	3.6×10^{-5} (explained below)
N	23,018	23,018	0
G (clusters)	3,114	3,114	0
K (regressors) / rank	24 / 24	24 / 24	0
Residual exposure SD (points)	0.428348	0.428348	none at six decimals
Unweighted counterpart β per 10	0.4147	0.4146971 (`m0008`)	none at four decimals

Note: Stored values are from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv` (rows `m0004` and `m0008`; β per 10 for `m0004` is 1.025553067200643 at full precision). Script 20 compared its estimate with six-decimal constants written into the script (β 1.025553, SE 0.260730), so the 6.7×10^{-8} in the β row measures

that rounding, not a disagreement between estimators; the full-precision comparison for this cell, made by `21_validate_all_primary_cells.py`, is 2.4×10^{-14} (pass-3 audit record). The independent rank is that of the regressor matrix before the fixed-effect transformation; the stored rank is that of the weighted within-transformed matrix. Independent values are carried from the RESTRICTED execution log; where only rounded values are recorded, agreement is stated at that precision.

Full execution log and comparison JSON: `data/processed/estimation/pooled_independent_validation_20261007T090003Z/` (RESTRICTED; aggregate figures only, reproduced above).

Which of the two changes explains the 1.385-versus-1.026 gap (2026-10-09 diagnostic). Both changes were made together in the original attempt, and that attempt's code was never committed to this repository, so the gap could not at first be decomposed. A dedicated diagnostic (`scripts/historical_wash/25_diagnose_birthyear_coding_and_age_control.py`) separates them on a reconstruction of the described design. It compares the two birth-year schemes on the transformed design the estimator actually uses, the weighted within-cluster-demeaned matrix, not on the raw matrix, because raw-space comparisons alone cannot show whether a fitted coefficient must be unaffected. **The exposure variable (E10, exposure in units of 10 percentage points) is never used to construct any of the comparisons below.** An earlier version of the diagnostic residualized the birth-year blocks against a set that included E10 itself before comparing ranks. That is circular: residualizing against E10 can remove exactly the part of a block that would otherwise affect E10's own coefficient. The corrected version keeps E10 out of every column-space comparison and instead checks directly that E10's own residual values do not depend on the choice of scheme. The script's tolerances are a rank tolerance of 10^{-8} relative to the largest singular value, a weighted column-norm threshold of 10^{-10} for treating a demeaned column as absorbed, and a relative tolerance of 10^{-8} for "the same residual vector".

Raw design matrix. The single-pooled-reference birth-year scheme does change the raw design matrix: it has 13 birth-year dummy columns against the country-local scheme's 12. Nigeria's observed birth-year range (2013–2017) does not include the single global reference year (2009, the earliest birth year observed anywhere in the pooled sample), so Nigeria keeps a dummy for its own earliest year, which the country-local scheme drops as its reference category. This extra column survives the cluster-fixed-effect demeaning on its own; it is not itself absorbed.

Step 1: direct comparison, before E10 or any other covariate is involved. Each scheme's birth-year block is demeaned (weighted within-cluster demeaning, the transformation that `PanelOLS`'s entity effects use) and the weighted blocks' ranks are compared directly: local 12 (of 12 surviving columns), pooled single reference 12 (of 13 surviving columns, one of

which is already redundant at this stage), combined 12. The two demeaned blocks span **equal** subspaces before any residualization.

Step 2: residualized against the shared nuisance controls only. The blocks are residualized against age in months, maternal age and the demographic dummies, with E10 explicitly excluded from the projection, and the check is run separately with and without age in months. The spans are equal in both cases (rank 12/12/12 throughout).

Step 3: E10's own residual, compared directly. For each scheme, the complete nuisance specification is built (the absorbed cluster fixed effect, the shared controls and that scheme's own birth-year block) and E10 is residualized against it. The two weighted residual-exposure vectors differ by at most 1.73×10^{-15} in absolute terms; divided by the weighted norm of the residual vector, that maximum is 2.23×10^{-16}. The vectors are the same, which is confirmed elementwise rather than inferred from rank alone.

Step 4: independent cross-check with the explicit weighted Frisch–Waugh–Lovell formula

$$\hat{\beta} = \frac{e' W\tilde{y}}{e' We},$$

where W is the diagonal matrix of model weights, e is the demeaned E10 residualized on that scheme's nuisance columns, and $\tilde{y}$ is the demeaned HAZ residualized on the same columns. This β is computed from first principles, not read from the output of `PanelOLS`, and it agrees with the coefficient that `PanelOLS` itself fits to within 4.4×10^{-16} in all four scheme and age-control combinations.

Step 5: mechanical explanation of the redundant column, without reference to E10. In the pooled-single-reference scheme, Nigeria's five own-year dummy columns sum to Nigeria's own raw country indicator, confirmed elementwise across all 23,018 observations (absolute tolerance 10^{-12} for these 0/1 columns). That indicator demeans to exactly 0.0: every DHS cluster belongs to exactly one country, so a column that is constant within every cluster is absorbed by the cluster fixed effect. This is the mechanism behind the equal-span result above.

Steps 1–3 establish directly, not by inference from a single rank comparison, that the two birth-year schemes are equivalent nuisance specifications for the exposure coefficient. Step 4 confirms the resulting coefficient by an independent computation, and Step 5 explains the mechanism. The fitted values agree with this result at floating-point precision. With the age-in-months control included, the exposure coefficient is 1.0255530672006674 (local) against 1.0255530672006667 (pooled single reference), an absolute difference of $6.661338147750939\times10^{-16}$. With the control excluded, it is 1.3852855197991372 against 1.3852855197991374, an absolute difference of $2.220446049250313\times10^{-16}$. These differences are three and one units in the last place of a double-precision number near 1

(machine epsilon 2.2×10^{-16}). They are reported as independently observed numerical evidence consistent with the algebraic result, not merely inferred from it.

The age-in-months control, by contrast, accounts for the gap on this reconstruction: 1.0255530672006674 with it and 1.3852855197991372 without it, under *either* birth-year scheme. The first value matches the stored value (1.025553). The second approximately reproduces the documented early-attempt value ("1.385"). On this reconstruction, therefore, the age-in-months omission explains the 1.385-versus-1.026 gap. The birth-year reference coding does change the raw design matrix, and this is not dismissed as a non-issue, but it does not affect the fitted exposure coefficient, because the two schemes carry the same information after cluster-fixed-effect absorption. This is the diagnostic's finding on a reconstruction of the described design. It is not a claim about what the specific, unpreserved original code did, and it is not an exact recovery of that code's reported output. The full tolerances and the complete five-step evidence chain are in the diagnostic's own execution log (`data/processed/estimation/birthyear_coding_diagnostic_20261009T081301Z/`, RESTRICTED); the values above are also recorded in the script's docstring.

The SE difference in Table B.2 is fully explained; it is not residual numerical noise. The declared small-sample correction of `16_estimate_preliminary.py` is $(G/(G-1)) \times ((N-1)/(N-K))$, applied to the raw cluster-robust sandwich, with confidence intervals and *p*-values from a Student *t* distribution on $G-1$ degrees of freedom. Here K counts the non-fixed-effect regressors retained (24 for the pooled cell); the absorbed cluster effects are not counted (function `fit()`, lines 248, 251 and 264–265 at commit `c58fc47`). The **original, incomplete** call in `20_validate_pooled_estimate.py` and `21_validate_all_primary_cells.py`, `PanelOLS.fit(cov_type="clustered", cluster_entity=True, auto_df=True)`, applied only one of the two factors that this correction requires. `auto_df=True` correctly sets the same-group, no-double-penalty degrees-of-freedom convention: with the entity effect nested in the clusters, `_determine_df_adjustment` in `panel/model.py` sets the extra degrees of freedom to zero (confirmed in the `linearmodels` 7.0 source). `debiased=True`, the library's own default, applies the scale $N/(N-K)$. The original call did not pass the *separate*, opt-in `group_debias=True` setting, so it never applied the library's cluster-count adjustment $(G/(G-1)) \times ((N-1)/N)$ (`group_debias_coefficient` in `shared/covariance.py`). With that setting added, the two factors multiply out to the declared correction:

$$\frac{N}{N-K} \times \left(\frac{G}{G-1} \times \frac{N-1}{N} \right) = \frac{G}{G-1} \times \frac{N-1}{N-K}.$$

The difference between the two pipelines is therefore a covariance-setting discrepancy in the validation scripts, not a different formula and not floating-point noise. `scripts/historical_wash/24_reconcile_inference_corrections.py` refits all 24 primary cells with `group_debias=True` added, using sample-construction code copied from script 21. For every one of the 24 cells, the pass-4 audit record reports a maximum absolute SE difference of 0.0 against the stored value at its displayed eight-decimal precision. A direct algebraic check (the uncorrected sandwich SE multiplied by the manually computed combined scalar, compared with the library's own corrected SE) leaves a residual of 0.0 in all 24 cells.

A reconciled SE does not by itself reconcile confidence intervals and *p*-values, because these also depend on the reference distribution. `linearmodels` computes its native intervals and *p*-values from a *t* distribution on `df_resid` = $N - K - G$, which counts every absorbed entity effect (1,571 to 19,918 in these cells, against $G - 1$ of 352 to 3,114). Script 24 therefore computes confidence intervals and *p*-values explicitly from the corrected SE with Student $t(G-1)$, the stored pipeline's reference distribution, and does not use the library's native values. Against the stored values, the maximum absolute differences over the 24 cells are 1.4×10^{-13} (lower CI bound), 7.6×10^{-14} (upper CI bound) and 1.2×10^{-13} (*p*), all within the script's "exact" tolerance of 10^{-9}. The native `df_resid`-based values, shown for contrast only, differ by up to 0.0065 in a CI bound and 0.00069 in *p* in three of the six Ghana cells, which have the smallest cluster count ($G = 353$). Full detail is in `docs/provenance/independent_audit_v1_2026-10-07/09_update_2026-10-07_pass4.md`; the per-cell table is RESTRICTED. None of this changes the stored, authoritative SE in the output of `16_estimate_preliminary.py`. That estimator's formula was correct throughout; what was corrected is the covariance configuration of the *validation* scripts, not the pipeline being validated. The SE in Table B.2 is kept as originally run, and the CIs and *p*-values printed by script 20 itself, which use the native reference, are not reported here.

Coverage across the 291 historical models. The verification described above covers the following, and no more:

1. All 24 primary established-date cells (2 products × 3 outcomes × Ghana, Kenya, Nigeria and pooled) were re-estimated with a second library on separately rebuilt samples: β, SE, N and G by script 21, then SE, CI and *p* under the corrected covariance setting by script 24.
2. Two further cells, both unweighted counterparts, were re-estimated individually: `m0005` by script 17 and `m0008` by script 20.

3. The remaining 265 models were not re-estimated with a second implementation. They were checked only internally: by the estimator's own absorbed-column and rank safeguards and by the same-function verification rerun described in Section B.4, which reproduces the output but is not an independent implementation.

What this verification establishes, and what it does not. It establishes that the reported pooled coefficient and its standard error are computationally correct: they are the numbers that the documented specification produces when it is implemented correctly in a second, independent library with an equivalent covariance configuration. It does not establish that the specification identifies a causal effect. It does not show that the samples represent the populations of the surveyed countries, that the results hold outside them, that the weighting and selection limitations are resolved, or that the result is robust to the number of cells tested; no multiplicity adjustment is applied. Nor does it place this cell in a higher *reporting* tier than its sibling historical models, all of which share the same unresolved documentation issues (Section 6; `docs/provenance/historical_wash_issue_register_v1_2026-10-07.md`).

B.3 Design checks applied to all 24 primary historical models

The following were checked programmatically for all 24 primary models (`scripts/historical_wash/19_check_model_design.py`): record-key uniqueness; the cluster-to-survey mapping (one survey per cluster); country weight totals equal to N/M within each model's own estimation sample (relative tolerance 10^{-6}); full design-matrix rank, meaning that the rank recorded by the estimator equals the number of retained regressors; and the absence of singleton clusters in the final estimation data. All checks pass. The per-model results are public (`docs/provenance/historical_wash_preliminary_estimates_v1/design_checks_per_model_PRELIMINARY_AGGREGATE.csv`, 24 rows); the summary file is RESTRICTED (`data/processed/estimation/run_20261006T185646Z/design_checks_20261006T190020Z/design_checks_summary.json`).

M is the number of countries in that particular model: $M = 1$ for the 18 country-specific models and $M = 3$ for the 6 established-date pooled models, not $M = 3$ for all 24. The historical scripts set each country's weight total to N/M; the baseline pooled models set each country's total to 1 (Section 4.5). The two differ by the common constant N/M, which changes neither the weighted least-squares coefficients nor the cluster-robust variance: multiplying every weight by the same constant scales the bread of the sandwich by its inverse

and the meat by its square, and the two cancel. Equal country weight totals do not mean equal influence on the pooled slope, which depends on each country's residual exposure variation (Section 4.5). In the design-check script itself, the weight check is described as "N / K" (comment and output field `country_weight_totals_equal_N_over_K`), with K meaning the number of countries; `equal_country_weights()` in `16_estimate_preliminary.py` uses K in the same sense. This appendix writes *M* for that count. Elsewhere in the design-check script, the variable `K` is the regressor count, which is also this appendix's *K*.

B.4 Estimator safeguard: the exposure-absorption and rank-deficiency defect, and its resolution (2026-10-10)

An external independent audit (`Independent_Thesis_Audit_2026-10-10.md`, which is not in this repository; its finding R05 and the resolution are recorded in `docs/provenance/independent_audit_resolution_v1_2026-10-10.md`) identified a genuine code defect in the `fit()` function of `scripts/historical_wash/16_estimate_preliminary.py`. The function's absorbed-column filter dropped any column with zero within-cluster variation after weighted demeaning, but the exposure coefficient and its variance were then read by **fixed array position** (`beta[0]`, `V[0,0]`), not by the retained variable's name. Suppose the exposure variable *E* itself were absorbed by this filter. The function's earlier check on the raw standard deviation would not catch this, because it tests *raw* (between-cluster) variation, not *within-cluster* variation. The next retained column would then silently occupy position 0 after the drop, and its coefficient would be reported as if it were *E*'s. Separately, the function computed a design-matrix rank but never compared it with the number of retained columns before using a pseudoinverse to solve for the coefficients. A rank-deficient retained design (for example, from an accidental exact collinearity between the exposure and a retained control) would therefore have returned a minimum-norm pseudoinverse solution silently, instead of flagging that the coefficient is not identified.

Fix. The function now (a) checks explicitly that `"E"` survives the absorbed-column filter and refuses to proceed if it does not (`status: "non_estimable_exposure_absorbed"`); when *E* survives, its coefficient and variance are retrieved by its retained name and index rather than by position. It also (b) checks that the rank of the retained design matrix equals its column count before computing coefficients, and refuses (`status: "non_estimable_rank_deficient"`) rather than returning a pseudoinverse-based

estimate if not. Neither change alters the computation for an estimable, full-rank case with *E* retained. The ordinary path recovers the identical coefficient, standard error and every other reported statistic, because *E* is always constructed as the first design-matrix column and so still occupies position 0 whenever it is retained.

Targeted tests are tracked at `scripts/historical_wash/26_test_fit_safeguard.py`. They use synthetic data only, run against the actual patched function through a direct import (not a reimplementation), and are re-run with `python 26_test_fit_safeguard.py` from that directory. The four cases are: (a) an ordinary estimable exposure, with genuine between- and within-cluster variation, which is estimated normally and recovers the simulated coefficient; (b) an exposure that varies between clusters but is constant within each cluster, the exact failure mode described above, which is correctly refused as `non_estimable_exposure_absorbed` with no `beta_per_point`, `se_CRV1` or `p_value` returned; (c) an exposure made exactly collinear with a retained nuisance control after demeaning, correctly refused as `non_estimable_rank_deficient`; and (d) a directly rank-deficient retained design created by a duplicated dummy block, correctly refused as `non_estimable_rank_deficient`. All four tests pass, according to the project's recorded run; they were not re-run in preparing this appendix.

Verification that no published model was affected. The patched script was re-run in full against the actual restricted inputs (`data/processed/analysis_samples/run_20261006T184744Z/` and `data/processed/child_exposure/run_20261006T183803Z/`, confirmed by SHA-256 to be bit-identical to the inputs of the original stored run). It wrote a fresh 291-model output to a new timestamped folder (`data/processed/estimation/run_20261010T140944Z/`, RESTRICTED) and left the original `run_20261006T185646Z/` unchanged. This rerun uses the same estimator function and the same inputs, so it shows that the patch left the output unchanged; it is not an independent implementation. A tracked comparison script, `scripts/historical_wash/27_compare_fit_safeguard_verification_run.py` (paths passed as `--original` and `--verification`; no restricted path is hard-coded as a requirement), produced a public, machine-readable comparison summary. All 291 model IDs are present in both runs, with no duplicates. All six specification fields (`family`, `product`, `outcome`, `sample_group`, `window`, `weighted`) match in every row, and both runs' `status` column reads `estimated` for all 291 rows. The maximum absolute difference is exactly 0.0 in each of the 15 numeric columns checked: the coefficient, SE and both CI bounds at both the per-point and the per-10-point scale, the *p*-value, `E_`

`within_sd`, `E_residual_sd_after_controls`, `N`, `G_clusters`, `K_regressors` and `rank`. This is the exact list, not an unenumerated "every column". The two runs' output CSV files are also byte-for-byte identical (identical SHA-256 hash), a stronger result than the per-column check requires.

Same-day correction (2026-10-10). The comparison script's overall-success flag was found not to account for a value that is finite on one side and NaN or infinite on the other, for duplicate IDs, for missing required columns, or for a NaN-versus-infinity difference. All of these are now fixed and covered by synthetic regression tests (`scripts/historical_wash/28_test_compare_fit_safeguard_run.py`). Re-running the corrected, stricter comparison on the same two runs reaches the same conclusion (`docs/provenance/fit_safeguard_verification_comparison_v2_2026-10-10.json`); the original, pre-correction summary is kept at `_v1_2026-10-10.json` for the record. Together, these results confirm directly, rather than by inference from the earlier design checks, that none of the 291 published historical models was affected by this defect: in every one, *E* was retained and the design matrix had full rank.

Appendix C — Historical exposure-assignment diagnostics and the Ethiopia provisional analysis

This appendix holds exploratory material and is consistent with Section 5.9. Nothing in it is part of the established-date pooled headline result, and none of it should be cited as such. It has two parts. Sections C.1–C.3 cover Ethiopia 2016, which is analysed only as a separately labelled provisional sensitivity outside the established-date pooled sample of Ghana 2014, Kenya 2014 and Nigeria 2018. Section C.4 summarises the spatial-extraction rule and selected sensitivity results of the established-date family.

The reporting status of each part follows the final reporting map (`docs/provenance/final_empirical_package_v3_2026-10-07/01_final_reporting_map.md`). Ethiopia candidate C1 is **invalid and superseded**. Candidates C2 and C3 are **provisional and exploratory**. The rows of Table C.3 belong to the **sensitivity family** of the established-date models. Like every historical estimate in this thesis, they are conditional associations among eligible measured children, not causal effects and not population estimates.

C.1 Ethiopia's three interview-date candidates

For Ethiopia 2016, the IPUMS-DHS extract provides no Gregorian interview date. It provides the interview year, month and day in the Ethiopian calendar and a century-day code for the interview date (`docs/provenance/historical_wash_milestone_2026-10-06.md`, section 4). The child's birth date is a different field: the Gregorian birth century-month code that IPUMS supplies already converted, at month precision (`KIDDOBCMC`). All three candidates share that birth date. The interview-date candidates therefore differ only in the interview date.

The interview date and the birth date enter the historical construction at different points (`scripts/historical_wash/12_construct_exposure_windows.py`, `14_build_analysis_samples.py`, `16_estimate_preliminary.py`):

1. **Birth date.** Each window's months, and hence its annual weights and its exposure *value*, are placed from the birth century-month code alone. The birth year, and with it the survey-qualified birth-year terms, is identical under all three candidates for all 10,641 Ethiopian children.
2. **Interview date.** The candidate interview date decides whether a child's dates are valid at all and whether the birth precedes the interview; both conditions apply to every

window. It also decides whether a postnatal window has been completed by the interview (the post-12-month and post-24-month windows only), and it sets the descriptive flag for an annual value that covers months after the interview.

3. **Age.** The under-five rule and the age control in the Ethiopia models use the reported completed age in months (`KIDCURAGEMO`), which the extract supplies for all 10,641 Ethiopian children (`docs/provenance/historical_wash_analysis_samples_v1/age_rule_reconciliation_AGGREGATE.csv`). Neither depends on the candidate. A calendar age computed from the candidate interview date exists in the data but is used only in the calendar-age sensitivity, which is estimated for the established-date samples and not for Ethiopia.

The three candidate conversions of the interview date are (`scripts/historical_wash/06_ethiopia_calendar_audit.py`):

- **C1** ("triple": 30-day Ethiopian-calendar arithmetic applied directly to the delivered year, month and day fields). **Invalid and superseded.** The delivered Ethiopian day and month fields are not original 30-day Ethiopian dates: the day value 31 occurs in months that the Ethiopian calendar defines as having 30 days. Direct 30-day arithmetic is therefore invalid for these fields. The milestone record that states this was committed on 6 October 2026, before the historical outcome models were estimated, and the finding does not depend on any outcome result. C1 was nevertheless estimated as a provisional candidate in that run. It was reclassified as invalid and superseded on 7 October 2026 (`docs/provenance/historical_wash_ethiopia_candidate_status_v1_2026-10-07.md`, section 1). Under this method, 275 of the 10,641 Ethiopian children have a day value that cannot be converted at all, so C1 gives them no date; the conversions for the other children use the same invalid logic. C1 is excluded from every reporting table in this thesis. Its earlier numerical agreement with C2 and C3 is not treated as evidence for either of them, because a method already known to be invalid cannot corroborate another one.
- **C2** ("century-day route"): the delivered century-day code read with the origin 1 Meskerem 1900 = 12 September 1907. Provisional; not independently validated.
- **C3** ("century-day route", anchored): 1 Meskerem 2008 = 12 September 2015, with the delivered month and day fields counted in Gregorian cumulative month lengths. Provisional; not independently validated.

The candidate-status record (section 2) names the weakness that C2 and C3 share, quoting the milestone record: the century-day code follows a month-length structure that

differs from a 30-day count, and its origin and the source of a constant two-day offset are not established. Neither candidate has been independently validated. A support inquiry about the calendar question was drafted but, according to the dated records of 6 October 2026, not sent (milestone record; `docs/provenance/historical_wash_decisions_2026-10-06.md`); this repository contains no record that it was sent or answered.

C.2 How closely C2 and C3 agree on window eligibility

Table C.1 — Ethiopia 2016: children eligible for each exposure window under candidates C2 and C3

Window	C2 eligible	C3 eligible	Disagree
Birth-year	9,862	9,862	0
Prenatal (9 months)	9,862	9,862	0
Post-12-month	7,930	7,922	8
Post-24-month	5,974	5,969	5

Note. Counts of children, out of 10,641, whose exposure for the window is eligible under the primary complete-coverage construction, before any outcome-specific exclusion; they are not model samples. "Disagree" counts children who are eligible under one candidate and not under the other. Each row is a different window for largely the same children, so the rows must not be added. Evidence: the candidate-status record (`historical_wash_ethiopia_candidate_status_v1_2026-10-07.md`, section 3) reports these comparisons from the restricted analysis file. It states 9,862 for both candidates in the birth-year and prenatal windows and the disagreement counts 0, 0, 8 and 5, but gives the postnatal counts only approximately (about 7,930 and about 5,970). The exact postnatal counts are carried from the canonical appendix; no public output contains them. In both postnatal rows the difference between the two counts equals the number of disagreements.

Table C.1 reflects how the interview date enters eligibility (Section C.1). In the birth-year and prenatal windows, the interview date matters only through date validity and the requirement that birth precede the interview, and C2 and C3 classify every child in the same way. In the two postnatal windows, the completion check compares the window's end with the interview month, and the two candidates place a few borderline children on different sides of it: 8 children in the post-12-month window and 5 in the post-24-month window. The stored analysis samples show the same pattern. For water and HAZ, the birth-year and prenatal samples are 8,250 children under both candidates, while the post-12-month samples are 6,580 under C2 and 6,573 under C3 before singleton clusters are dropped, and 6,568 and 6,561 in the estimated models (`docs/provenance/historical_wash_analysis_samples_v1/design_diagnostics_by_sample_window_AGGREGATE.csv`; `model_coefficients_PRELIMINARY_AGGREGATE.csv`, rows `m0045` and `m0046`).

The comparison with C1 is narrower. For the birth-year window, the stored aggregate

shows that C1 differs from C2 and C3 for exactly 275 children, all of them the children without a C1 date; no candidate places any birth after the interview (`docs/provenance/historical_wash_analysis_samples_v1/ethiopia_birthyear_attribution_AGGREGATE.json`). This stored comparison covers the birth-year window only.

These counts support two narrow statements. First, the choice between C2 and C3 changes the window classification of very few children. Second, this is **not** evidence that either candidate's absolute date is correct. The two candidates share the unexplained recode structure described in Section C.1, so their agreement is not independent validation. Nor is the difference between them a confidence interval, an uncertainty bound or a guaranteed range for the true interview date: both could be wrong in the same direction.

A valid Gregorian interview date is still required for every child in every window, because it decides whether the child is validly dated and whether the birth precedes the interview (Section 3.7). This requirement is distinct from the postnatal-window **completion** check, as Section 3.7 and the construction script state. The completion check asks whether all months of the window had ended by the interview month, and it applies only to the post-12-month and post-24-month windows. The prenatal window ends before birth and needs no such check. The birth-year window has no completion check at all: a child born early in a year and interviewed before that year ends is still assigned that year's full annual value. Eligibility in the birth-year window still requires valid dates, a birth year inside the 2000–2017 raster span and complete spatial coverage. In any window, the annual value used for the interview's own calendar year can cover months after the interview; such cases are flagged and not excluded (Section 3.7). The candidate-status record (section 4) describes the completion rule as applying to every window, including the birth year, citing the design draft; the implemented construction does not do this, and Section 3.7 and this appendix follow the implementation.

C.3 Ethiopia provisional coefficients (water HAZ, sanitation HAZ, post-12-month window)

Table C.2 — Ethiopia 2016 provisional HAZ coefficients under candidates C2 and C3 (post-12-month window, per 10 percentage points)

Candidate	Water β per 10	Sanitation β per 10
C2	−0.39	−0.45
C3	−0.39	−0.45

Note. Stored values from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`, family `sens_ethiopia_PROVISIONAL`, outcome HAZ, window `post_12m`. Water (`W_IMP`): C2 `m0045`, −0.39 [−0.84, 0.06], p = .086, N = 6,568; C3 `m0046`, −0.39 [−0.84, 0.06], p = .088, N = 6,561. Sanitation (`S_IMP`): C2 `m0186`, −0.45 [−1.32, 0.41], p = .303, N = 6,568; C3 `m0187`, −0.45 [−1.31, 0.42], p = .312, N = 6,561. G = 586 clusters in all four models. Specification as for the established-date models (Section 4.6): cluster fixed effects, primary buffer with complete coverage, `PERWEIGHT` normalised within the estimation sample, reported age, wealth excluded. Standard errors are CRV1 clustered by DHS cluster; confidence intervals and p-values use a Student t distribution with G − 1 degrees of freedom. C1 estimates exist in the same file but are not shown, because C1 is invalid.

All four confidence intervals include zero. The estimates are reported here, and cross-referenced in Section 5.9, as an exploratory sensitivity with no bearing on the established-date pooled results in Section 5.6. Ethiopia is not pooled with Ghana, Kenya and Nigeria under any established-date label. The near-identical C2 and C3 coefficients follow from the near-identical samples in Section C.2 and do not validate either date. The gap between the two estimates is not a measure of how far either lies from an estimate based on the true interview dates.

C.4 Spatial-extraction and sensitivity diagnostics (established-date models, summarised)

Each result row of Table C.3 changes one element of the main pooled specification and compares the pooled water–HAZ coefficient with the main estimate. The main estimate is model `m0004`: Ghana 2014, Kenya 2014 and Nigeria 2018 pooled, improved water (`W_IMP`), HAZ, post-12-month window, primary buffer with complete coverage, `PERWEIGHT` with equal country totals, reported age where delivered, wealth excluded, coefficient per 10 percentage points. Everything else is held fixed. Each sensitivity model, however, is estimated on its own eligible sample, so a difference from the main estimate can reflect a change in the children included as well as the change in specification.

Table C.3 — Spatial-extraction rule and sensitivity of the pooled established-date water–HAZ coefficient (per 10 percentage points)

Diagnostic	Result
Buffer coverage-completeness tolerance	Within 10^{-3} of full buffer area (valid-area fraction $\geq 1 - 10^{-3}$)
Spatial-method sensitivity (containing-pixel vs. buffer average)	Pooled water–HAZ: 0.68 vs. main 1.03
Rural 10 km buffer sensitivity (urban unchanged)	Pooled water–HAZ: 1.01 vs. main 1.03
Stricter postnatal completion rule (+1 month)	Pooled water–HAZ: 0.95 vs. main 1.03
Calendar-month age construction throughout	Pooled water–HAZ: 1.03 (materially unchanged)
Ghana excluded (DHS/IHME overlap; Kenya's water overlap remains)	Kenya and Nigeria pooled water–HAZ: 0.74 vs. three-country main 1.03

Note. Stored values from `docs/provenance/historical_wash_preliminary_estimates_v1/model_coefficients_PRELIMINARY_AGGREGATE.csv`, all HAZ, `W_IMP`, post-12-month window, weighted. Main `m0004` (primary): 1.03 [0.51, 1.54], $p < .001$, N = 23,018, G = 3,114. Containing pixel `m0028`: 0.68 [0.26, 1.10], $p = .002$, N = 23,642, G = 3,196. Rural 10 km `m0032`: 1.01 [0.42, 1.60], $p = .001$, N = 22,521, G = 3,062. Stricter completion `m0036`: 0.95 [0.43, 1.47], $p < .001$, N = 22,639, G = 3,113. Calendar age `m0044`: 1.03 [0.52, 1.53], $p < .001$, N = 23,018, G = 3,114. Ghana excluded `o003` (family `sens_overlap_excl_GH_water`, Kenya and Nigeria only): 0.74 [0.24, 1.24], $p = .004$, N = 21,078, G = 2,761. Inference as in Table C.2: CRV1 clustered by DHS cluster, $t(G - 1)$ confidence intervals and p-values.

Coverage rule and spatial assignment. The first row states a rule, not a result. For each cluster and year, the extraction computes the exact area of overlap between a disk of the stated radius and each raster pixel, after projecting the pixel's corners into a local azimuthal equidistant projection (Section 3.6). Pixels outside the raster, NoData pixels and values outside 0–100 are excluded from the valid area; they are never set to zero, and the buffer mean is taken over the valid area only. An annual value counts as complete when the valid area is at least $1 - 10^{-3}$ of the buffer's total area (`scripts/historical_wash/10_spatial_extraction_v3.py`, `FULL_TOL`). A window is used only if every year it touches has a complete value within 2000–2017; nothing is extrapolated or filled. The primary buffer is 2 km for urban and 5 km for rural clusters. Partial-coverage values are kept only as a diagnostic status and enter none of the 291 stored models. The buffer is centred on the DHS-displaced cluster location, so it does not recover the true location. It is an area average of modelled coverage, not a population-weighted coverage estimate.

Containing pixel. This row replaces the buffer average with the value of the single pixel that contains the displaced cluster point. NoData gives a missing value, and no nearest-cell value is substituted. Because a valid value is needed only for that pixel, the eligible sample differs from the main one (N = 23,642 against 23,018). The move from 1.03 to 0.68 therefore

reflects both the assignment rule and the sample.

Rural 10 km. Rural children are assigned the 10 km buffer average; urban children keep the 2 km primary buffer. The sample again differs (N = 22,521). A larger rural radius is a sensitivity to the size of the averaging area. It does not validate the DHS displacement probabilities or recover true locations.

Stricter completion. The postnatal completion rule is tightened from "interview month at least L months after the birth month" to at least L + 1 months, because the birth day is unknown. It applies only to the post-12-month and post-24-month windows; the table shows the post-12-month case. The estimation sample falls from 23,018 to 22,639.

Calendar age. The age control uses the calendar-month difference between interview and birth for all children instead of the reported completed age where it is delivered. In the established-date family, reported age is delivered only for Nigeria; Ghana and Kenya use calendar age in both versions. The sample is unchanged (N = 23,018), and the coefficient moves only in the third decimal (1.027 against 1.026).

Ghana excluded. Ghana's 2014 DHS round is an input to the IHME model for both products, and Kenya's 2014 round is an input for water (`docs/provenance/historical_wash_issue_register_v1_2026-10-07.md`, item 7). Removing Ghana leaves a two-country model pooled over Kenya and Nigeria (N = 21,078), not a three-country pooled model, and Kenya's water overlap remains in it. This row therefore shows the effect of removing Ghana's overlap, not an estimate free of DHS/IHME overlap. The stored file also contains a water model excluding both Ghana and Kenya; it leaves Nigeria alone and is a country-specific estimate, not a pooled one.

Weighting. Every row uses `PERWEIGHT` with equal country totals within its own estimation sample. Equal country totals do not give the countries equal influence on the pooled slope, which also depends on each country's residual exposure variation. The unweighted counterpart of the main model (0.41) is reported in Section 5.6 and Appendix B. Neither the weighted nor the unweighted version corrects for selection into anthropometric measurement. The window and wealth sensitivities of the same family are shown in Section 5.7 and Section 5.8.

No sensitivity cell was selected or suppressed because of its significance. The sensitivity families were listed in the estimation specification (`docs/provenance/historical_wash_analysis_specification_v1_2026-10-06.md`, section 8). That record states that no coefficient had yet been estimated, and it was committed before the recorded time of the 291-model estimation run. The families are reported together in the methods note (`docs/provenance/historical_wash_preliminary_estimation_methods_2026-10-06.md`, section 7), and every

one of the 291 models, including all outcomes and both products, is in the stored coefficient file. Table C.3 shows the pooled water–HAZ cells only.

What these diagnostics do not establish. All 291 historical models were estimated by `16_estimate_preliminary.py`. The 24 primary established-date cells, including the main model `m0004`, were independently reproduced with a second implementation, and two unweighted counterparts (`m0005`, `m0008`) were checked individually (Appendix B). The remaining 265 models, including every other row of Table C.3 and all 18 Ethiopia models, were not reproduced with a second implementation; they passed only internal checks. Agreement among these internally consistent diagnostics is not independent replication. Neither the reproduction nor the sensitivity results resolve causal identification, the undocumented weighting of the measured subsample, differential non-response among eligible children, the Nigeria 2018 measured-count discrepancy, the DHS/IHME input overlap, the use of current residence for early-life residence, coordinate displacement, the uncertainty of the modelled exposure surface, the uncertainty of Ethiopia's interview dates, or multiple testing, for which no adjustment has been made (Section 3.8, Section 5.6, Section 6). These remain qualifications of the research, not implementation errors.

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

Violations of this declaration may be considered an attempt at deception.

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Place, Date Signature